

VERIFICATION REPORT

SUMATRA MERANG PEATLAND PROJECT (SMPP)



AENOR INTERNACIONAL S.A.U.

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Summary

AENOR has carried out the verification of the Sumatra Merang Peatland Project (SMPP) under the VCS and CCB Programs. The project is a combination of Afforestation, Reforestation and Revegetation (ARR) project and Wetlands Restoration and Conservation (WRC) project implemented in South Sumatra province, Sumatra, Republic of Indonesia. The project aims to protect and restore 22,922 ha of peatland ecosystem, while building strong and supportive relationships with the broad range of stakeholders within the landscape. Project activities include forest patrolling, fire prevention and monitoring, canal blocking and hydrology management, reforestation and assisted natural regeneration, as well as development activities for communities neighboring the project area, such as education and health support or provision of access to clean water.

The project start date is 1 January 2016 and will be operational until 31 December 2062. The estimated net GHG emissions reductions and removals at validation for the 47-year project span were 62,912,755 tCO₂e, at an average of 1,338,569 tCO₂e/yr. The emissions reduction for the current monitoring period (January 1st, 2021 – December 31st, 2021) is 1,619,710 tCO₂e. This is not a grouped project.

The purpose of the verification was to determine the conformance of the project with respect to the VCS Version 4 and CCB Third Edition and the validated PD, and the assessment of the ex-post monitored anthropogenic GHG emissions reductions and/or removals that have occurred as a result of the project's activities. The scope of the verification was to assess the conformance of validated project, once implemented, with the VCS and CCB requirements and requirements in the validated PD. The process was performed through a combination of desk review, interviews, and communications with relevant personnel. This is the fourth verification event, corresponding to the monitoring period from 1 January 2021 to 31 December 2021.

During the verification 5 CLs and 7 CARs were raised. All these issues were appropriately closed by means of corrections, more clear explanations, and other supporting documents.

Once all issues detected were appropriately resolved, AENOR carried out this final verification report and deems with reasonable level of assurance that the project complies with all of the verification criteria. The assessment team has no restrictions or uncertainties with respect to the compliance of the project with the verification criteria, hence, the audit team concludes that the cumulative net GHG emissions reductions or removals of 1,619,710 tCO₂e over the monitoring period has been quantified in accordance with VCS rules. A buffer discount rate of 14% was applied, resulting 1,392,951 VCUs eligible for issuance. AENOR confirms that the project has achieved the Climate Gold and Biodiversity Gold distinctions for the verified monitoring period in accordance with the Third Edition CCB Standards.

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1 INTRODUCTION

1.1 Objective

The objective of the verification audit was to conduct an independent assessment of the project to determine:

- The extent to which methods and procedures, including monitoring procedures, have been implemented in accordance with the validated project description, including the monitoring plan.
- The extent to which GHG emission reductions and removals reported in the monitoring report are materially accurate.

1.2 Scope and Criteria

The scope of the verification included the review of the GHG project and implementation; physical infrastructure, activities, technologies, and processes of the GHG project; GHG sources, sinks and/or reservoirs; types of GHG's; and time periods covered. The project follows the framework an Agriculture, Forestry, and Other Land-Use (AFOLU) project. Project activities are categorized as a combination of Afforestation, Reforestation and Revegetation (ARR) and Wetlands Restoration and Conservation (WRC). Under the WRC component, the project activities relate to both the Conservation of Undrained and Partially Drained Peatland (CUPP) and Rewetting of Drained Peatland (RDP). The geographic verification scope is defined by the project boundary, the carbon reservoir types, management activities, inventory program, and contract periods.

The monitoring period for this verification is from 01 January 2021 to 31 December 2021.

The scope of this audit included a verification of the projects calculated reductions and removals with the Verified Carbon Standard requirements. In addition, the audit assessed the project with respect to the validated baseline scenarios presented in the PD and the fulfilment of the climate, community, and biodiversity criteria against the CCB Standard.

The scope was defined as follows:

- The project and its baseline scenarios;
- The physical infrastructure, activities, technologies and processes of the project;
- The GHG sources, sinks and/or reservoirs those are applicable to the project;
- The types of GHGs that are applicable to the project; and
- The project monitoring period
- The climate, community, and biodiversity benefits, including gold distinctions

Standard criteria: Criteria from the following documents were used to assess this project:

- VCS Program Guide, v4.2
- VCS Standard, v4.3
- Program Definitions, v4.2
- AFOLU Non-Permanence Risk Tool, v4.0
- Climate, Community & Biodiversity Standards, v3.1

- CCB Program Rules, v3.1

Unless otherwise indicated, the assessment was performed against the most recent version of the relevant VCS and CCB guidance documents.

1.3 Level of Assurance

The assessment was conducted to provide a reasonable level of assurance of conformance against the defined audit criteria and materiality thresholds within the audit scope. Based on the audit findings, a positive evaluation statement reasonably assures that the project GHG assertion is materially correct and is a fair representation of the GHG data and information.

The threshold for materiality with respect to the aggregate of errors, omissions, and misrepresentations relative to the total reported GHG emission reductions/removals was one percent (1%), as established for large projects by the VCS Standard.

All the revisions of the verification report before being submitted to the client were subjected to an independent internal technical review to confirm that all verification activities had been completed according to the pertinent AENOR instructions required. The technical review was performed by a technical reviewer(s) qualified in accordance with AENOR's qualification scheme for CDM/VCS validation and verification.

1.4 Summary Description of the Project

The project is an ARR and WRC project implemented in the implemented in South Sumatra province, Sumatra, Republic of Indonesia. It aims to protect and restore 22,922 ha of peatland ecosystem, while building strong and supportive relationships with the broad range of stakeholders within the landscape. Project activities include forest patrolling, fire prevention and monitoring, canal blocking and hydrology management, reforestation and assisted natural regeneration, as well as community development sub-activities. The Project Objective is to restore the ecological functions of the Merang Peatland through the use of sustainable land-use models that benefit critically endangered biodiversity and provide benefits for local communities. Throughout this period, the project has supported a daily forest patrol system, and with daily weather monitoring. The project continued to implement community development activities and other programs achieving positive impacts for stakeholders. During the monitoring period an additional 15 dams were successfully constructed, bringing the total to 203 permanent canal blocking dams. The specific objectives are:

- Climate Objectives
 - Credibly avoid peatland emissions through the protection and rehabilitation of the project site.
 - Restore the variety, integrity, and extent of forest types to their original pre-2007 fire state to rejuvenate the presence of valuable ecosystem services and peatland habitats.
 - Restore the hydrological integrity of the peatlands.

- Prevent fires and ensure the safety of local people through forest fire training, monitoring, and prevention procedures.
- Community Objectives
 - Introduce economic opportunities and sustainable livelihood opportunities, focused on community-directed investments with both immediate, short-term, and long-term strategies in the SMPP's neighboring communities.
 - Build stronger communities.
 - Ensure continued access and sustainable use of natural resources by communities and local stakeholders.
- Biodiversity Objectives
 - Increase populations of critically endangered flora and fauna by eliminating the causes of deforestation and degradation in the project site.
 - Establish ecological integrity with long-term ecosystem restoration.

The project start date is 1 January 2016 and will be operational until 31 December 2062. The estimated net GHG emissions reductions and removals at validation for the 47-year project span were 62,912,755 tCO₂e, at an average of 1,338,569 tCO₂e/yr. The emissions reduction for the current monitoring period (01-January-2021 to 31-December-2021) is 1,619,710 tCO₂e. This is not a grouped project.

2 VERIFICATION PROCESS

2.1 Audit Team Composition (*Rules 4.3.1*)

Name	Position in the team
Javier Cócera	Lead auditor
Daniel Bermejo	Auditor
Adrián Vidal	Auditor trainee
Waqar Ahmed	CCB expert
José Luis Fuentes	Technical reviewer

Javier Cócera is a forest engineer with a Master in forest management. He has developed his career focused to the forest management. Mainly he has been working through sustainability in two ways: the main one as forestry consultancy, developing forest management plans, working with GIS and LiDAR both in the field and the office and getting experience of the forest resources. The second one was developing environmental footprint projects and sustainability reports. Currently Javier is working in AENOR as auditor focused in AFOLU projects. Javier participated in courses about ISO lead auditing and have performed audits in projects in Europe, LATAM, Africa and Asia.

Daniel Bermejo is a Spanish forest engineer with a Master in sustainable finance. He started his career in private consultancy working on topics as climate risk analysis and TCFD risks, EU Taxonomy, development

of banking sustainable standards (agriculture, biodiversity and forestry) and environmental footprint projects and sustainability reports. He is currently working in AENOR as an auditor focused on AFOLU projects.

Adrian Vidal holds a Master's degree in Forest Engineering from the Technical University of Madrid, and a Postgraduate Diploma in Climate Change from the National University of Quilmes and the National University of Jujuy, with the support of UNEP. Prior to joining AENOR, he worked at the Basque Center for Climate Change (BC3) in the NDC ASPECTS project, carrying research in global governance, national policies, and modelling of Agriculture, Forestry and other Land Use (AFOLU) mitigation measures. He worked as an intern at the AFOLU Unit of the Transparency division of UNFCCC, providing support to the intergovernmental climate change process on issues related to land-use, land-use change and forestry (LULUCF), agriculture and REDD+. He also worked in urban forestry, landscape forest restoration and environmental consultancy, and collaborated in the Global Forest Survey project of FAO.

Waqar Ahmed is serving in the University of Karachi as a full-time faculty since 2006. Currently serving as an Assistant Professor, his duties include teaching, research and community service. He has also got experience in teaching as a visiting faculty in other universities like, University of Warsaw, Poland and Hamdard University. This demonstrates his strong communication skills. He has gained experience in research publications, with particular reference to marine pollution which is a part of his Ph.D. work, he has good command on the subject. He has also participated in various conferences at national and international level. Dr. Waqar completed various technical trainings in the country and abroad. He is also a certified trainer of the Integrated Coastal Management by IUCN. Dr. Waqar Ahmed has worked as a freelancer in various carbon assessment projects in Pakistan. He has worked as a technical expert in Delta Blue Carbon 1. He also has experience of providing consultation for implementation of biodiversity standards in CDC funded projects (following the IFC Guidelines on PS6, Biodiversity Management) in wind power plants in Pakistan. He has participated in various conferences at national and international level. Dr. Waqar completed various technical training in the country and abroad. He is also a certified trainer of Integrated Coastal Management by IUCN. He is also a member of IUCN Commission on Ecosystem Management. He has also served as a consultant in IUCN for various projects on biodiversity and environmental management. Dr. Waqar Ahmed has been serving in the University of Karachi as a full-time faculty since 2006. Currently serving as an Assistant Professor, his duties include teaching, research and community service. He has also got experience in teaching as a visiting faculty in other universities like, University of Warsaw, Poland and Hamdard University. With a Masters in Environmental Science and PhD which was on Ecology of Mangroves of Indus Delta he has good knowledge of ecology, mangroves and marine pollution. He has written 24 international research articles and is well versed with principles of ecology and biodiversity. Dr. Waqar teaches the subjects of Wildlife Ecology and Wetland Management, Climate Change, Marine Pollution etc. He has gained experience in research publications by writing 22 international research articles on several aspects of the environment. Dr. Waqar is fluent in English language in both spoken and written. He also learned the native Indonesian Language online and can communicate basic sentences of greetings and hello hi which is helpful to communicate with the stakeholder.

José Luis Fuentes is the manager of the Climate Change Unit of AENOR. He is a Forestry Engineer and has a master's in business administration and a Post-Graduate in Environmental Management. He has more than 15 years of experience in auditing, consulting and training activities related to environmental and carbon management projects. Jose Luis has actively participated in the audit of international sustainable development projects in several carbon schemes, such as the Clean Development Mechanisms (CDM), Verified Carbon Standard (VCS), Climate, Community and Biodiversity Standards (CCB), Gold Standard

(GS) and carbon footprints (ISO 14067 and ISO 14064). Jose Luis has extensive technical knowledge about the regulatory framework, policies and technical provisions emanating from the Paris Agreement, the Kyoto Protocol and the Conferences of the Parties.

2.2 Method and Criteria

The verification was performed through a combination of document review and interviews with relevant personnel, as discussed in Sections 2.3 through 2.5 of this report. At all times, the project was assessed for conformance to the criteria described in Section 1.2 of this report. As discussed in Section 2.6, findings were issued to ensure that the project was in full conformance to all requirements.

A project specific Verification and Sampling Plan was developed to guide the verification auditing process to ensure efficiency and effectiveness. The purpose of the Verification and Sampling Plan was to present a risk assessment for determining the nature and extent of verification procedures necessary to ensure the risk of auditing error was reduced to a reasonable level. The Verification & Sampling Plan methodology was derived from all items in our verification process stated above. Specifically, the sampling plan utilized the VCS guidance documents and ISO 14064-3. Any modifications applied to the Verification and Sampling plan were made based upon the conditions observed for monitoring in order to detect the processes with highest risk of material discrepancy.

The verification activities in which risks were assessed were the evaluations of the monitoring system (data flow, data control procedures, etc.) but mainly the quality of raw data as well as sources and the spreadsheet calculations. AENOR reproduced and verified 100% of sheets in the *MR_Tables_2021_20240119* and the other spreadsheets /36-47/ for the monitoring period 01 January 2021 – 31 December 2021 for the project area. The project boundary and land cover changes in the project area for the monitoring period were 100% checked using the GIS database /18-24/.

The carbon stock changes, and the land used classes in the project area were also 100% verified and crosschecked with validated values. For data provided for the reference region, AENOR carried out onsite samples of at least 5% of data since they had already been previously validated and posed a lower risk to the emissions reductions achieved by the project.

AENOR carried out a deep and meticulous review of the spreadsheets in order to verify the correct application of the methodology (formulae, equations.) and checked that data required for calculating the GHG removals and reductions were appropriately provided. Based on the assessment carried out, AENOR confirms with a reasonable level of assurance that the claimed emission reductions and removals are free from material errors, omissions, or misstatements.

AENOR confirms that sufficient evidence was presented for the reported net anthropogenic GHG emission reductions and removals and that there is a clear audit trail that contains the evidence and records that validate the stated figure in this verification report since:

- Sufficient evidence available: The project participant has provided the 100% of data used in the calculations to achieve the final amount of GHG emission reductions and removals reported.
- Nature of evidence: The raw data were collected from reliable sources. They are detailed in the project documents and have been provided to the verification team and the most relevant are appropriately detailed in the Appendix 1.

- Cross-checked evidence: AENOR cross-checked the collected information through interviews with stakeholders and reproducing calculations.

Hence, AENOR confirms that the stated figures in the monitoring report are correct and confirms that is able to certify net anthropogenic GHG reductions and removals based on verifiable and reliable evidence.

2.3 Document Review

A detailed review of all project documentation was conducted to ensure consistency with, and identify any deviation from VCS program requirements, CCB program requirements, the methodology (VM0007, v1.5), and the validated PD. Initial review focused on the Monitoring Report (MR) and included an examination of the project details, implementation status, data and parameters, and quantification of GHG emission reductions and removals. Documents reviewed included data from monitoring, economic analysis, maps and aerial images, fire specific monitoring data, deforestation and field reports, community development activity reports, SOPs, biomass and carbon calculation spreadsheets, and responses to Corrective Action Requests (CARs) and Clarifications (CLs).

The verification included a review of the validated PD and MR /9-11/, relative to the field conditions and interviews with project management staff, stakeholders and beneficiaries. Modifications to the Verification and Sampling plan were made based upon the conditions observed for monitoring in order to detect the processes with highest risk of material discrepancy.

The VCS AFOLU Non-Permanence Risk Tool was used by the Project Proponent to assess overall project risk. The VVB reviewed the Non-Permanence Risk Report /175/ provided with the verification supporting documentation and confirmed that the Project adheres to the requirements set out in the VCS AFOLU Non-Permanence Risk Tool. Each risk factor was thoroughly assessed for conformance. The final score was calculated to be 14%.

For a listing of all documents received from the client for this verification, please see Appendix 1.

2.4 Interviews

The AENOR's verification team composed of Waqar Ahmed conducted interviews with project developers; local stakeholders; and key personnel involved in the project activity, in order to confirm selected information and to resolve issues identified in the document review.

The field visit took place from 31- Aug-2022 to 06-Sep-2022 in which the technical expert visited the project area, interviewed key stakeholders, staff and other related experts, and also reviewed the monitoring report and supporting documents. The people interviewed were those directly affected or involved in the project activity and in some cases were just indirectly affected. Finally, some activities stated within the MR were also reviewed during the on-site visit and verified with the information provided.

Additional to site visit, meetings via teleconferences were carried during September 16th; in order to validate the baseline calculation and verify the processing data from satellite images. The scope of the verification was to assess the conformance of information in the monitoring report with the VCS and CCB requirements.

The complete list of interviewees is found in Appendix 2.

2.5 Site Inspections

The objectives of the on-site inspections performed were mainly to cross check the description provided in the monitoring report, related to the VCS and CCB requirement implemented by the proponent, including:

- Ensure that the geographic area of the project, as reported in the CCB-VCS-PD and the accompanying KML file, is in conformance with Section 3.10.2 of the VCS Standard;
- Perform a risk-based review of the project area to ensure that the project conforms to all other requirements of the VCS rules and the methodology.
- Observe the Project Proponent's evidence and collect and record data in order to assess whether data collection techniques conform to the monitoring plan and related documentation and to evaluate data quality control systems.
- Select samples of data and information for verification in order to meet a reasonable level of assurance and to meet the materiality requirements of the project, as required by Section 4.1.8 of the VCS Standard;
- Review of biomass plots and crosschecking of the verified information with the information gathered from the onsite visit. Plot sampling demonstration was done to replicate the plots which were used in measurements of the tree biomass. Sample plots were laid down as described in the Monitoring report. The number of plants in the plot were counted, DBH were measured using the measuring tape and height was measured using hypsometer.
- Perform a risk-based review of the project area to ensure that the project is in conformance the eligibility requirements of the VCS rules and the applicability conditions of the methodology;
- Interview local authorities to confirm that the project operates in accordance with current permits and authorizations and its relationship with local actors and communities.
- Interview the key personnel involved in the mentoring and observe monitoring practices.
- Verify activities such as planting, in the project zone.

Additional to the site inspection, meetings via teleconferences were carried with project representants and personnel in charge of carrying out the calculations, image processing, monitoring, additionality, and unique metrics report.

Based on the review of the provided documentation and considering the requirements of the standards used, the following areas were observed:

The following areas

- Project proponent, developers/management team, local team onsite
- Project activities conducted during the monitoring period (reporting, results and concordance with plan as per PD, etc.)
- Monitoring procedures, monitoring team and equipment and biomass plots layout and measurements
- Full review of operations and measurement records. Controls established to detect and correct any errors or omissions in monitoring parameters

- Forest conservation and implementation
- Project activities implemented
- Stakeholders' rights
- Stakeholders' identification and consultation, trainings/benefits/satisfaction
- Project conflicts, barriers or difficulties and grievances/complaints/conflicts
- Stakeholders' consultations, meetings and documents sharing
- Project communication
- Community, implementation of CCB community monitoring plan and benefits to the community
- Areas of High Conservation Value (HCV)
- Biodiversity and current problems associated. Implementation of CCB biodiversity monitoring plan and biodiversity conservation results including conservation of trigger species

2.6 Resolution of Findings

All documentation provided by the PP was assessed against the applicable version of the relevant VCS and CCB guidance document. Several clarification requests (CL) and corrective action requests (CAR) were raised and submitted to the PP, which addressed them either by providing to the audit team the requested information or by making the appropriate corrections. Updated versions of the documentation were submitted by the PP and the audit team reassessed them against the guidance documentation. This process was repeated iteratively until all CL and CAR were fully closed. Specifically, 5 CLs and 7 CARs were raised.

All findings issued by the AENOR audit team during the verification process have been closed. In accordance with Section 4.1.13 of the VCS Standard, all findings issued during the verification process and the inputs for their closure are described in Appendix 3 of this report.

2.6.1 Forward Action Requests

No forward action requests (FARs) were raised to the PP during the current verification process. No FARs were pending from the previous verification.

2.7 Eligibility for Validation Activities

AENOR has conducted the verification at the same time considering the paragraph 4.1.20 VCS standard. AENOR holds accreditation for validation and verification for the sectoral scope 14. Agriculture, Forestry, Land Use.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The verification team is not aware of project involvement in other forms of environmental credits from its activities. The project has not been registered, and is not seeking registration, under any other GHG programs. The project is already validated and verified under the VCS and CCB program.

3.2 Methodology Deviations

One methodology deviation has been assessed during this MP. The project proponent has made a minor adjustment from the methodology for CDM tool AR-AM-Tool-14 on the monitoring and measurement of carbon stock changes in the project scenario. This adjustment for carbon stock estimates in trees at a point of time now combines inventory and remote sensing methods to compliment the approach outlined in the methodology. This adjustment has resulted in increased accuracy by decreasing uncertainty to below 5%.

The only adjustment was the change to equation 13 of AR-AM-TOOL-14 which is replaced with equation 2.1 below, where B_{TREE} is replaced with $\sum_{i=1}^n B_{TREE}$, which is required for deriving the AGB estimates of each pixel for wall-to-wall biomass maps.

$$C_{TREE} = \frac{44}{12} \times CF_{TREE} \times \sum_{i=1}^n B_{TREE} \quad (2.1)$$

The tool VT0005 prescribes estimating aboveground live vegetation biomass by using remote sensing samples (e.g., airborne Light Detection and Ranging (LiDAR)) combined with in-situ inventory plots to extrapolate to the strata level (such as land use land cover classes) or project area of interest (AOI). the SMPP has utilized inventory plots (for year 2021) and airborne LiDAR samples (for year 2016) and complemented the LiDAR readings from a spaceborne database, modelled a relationship with wall-to-wall radar and optical satellite images, created a per-pixel vegetation height estimate map, then applied the model relating vegetation height to aboveground biomass (AGB), to produce the wall-to-wall per-pixel estimates of AGB.

This approach decreases uncertainty around the carbon stock estimation by calculating the total uncertainty as the relative error of combined uncertainties from modeled tree height and plot-based AGB, resulting in an uncertainty below 5% for this monitoring period. The project will monitor and model carbon stock changes in AGB in the project scenario through carbon stock mapping and analysis. The analysis will compare maps between monitoring periods. The calibrated model produces accurate wall-to-wall AGB maps with low uncertainty levels, when integrated with new remote sensing datasets. Uncertainties will be monitored and reported for each monitoring period to ensure that the highest uncertainty will not exceed the 10% threshold. The applicability conditions of VT005 are met because the project activity ARR requires the estimation of aboveground tree biomass. Remote sensing data were collected for the periods 2016 and 2021.

According to this methodology deviation, the VVB has performed technical meetings with the PP to assess the changes performed in the tool and the corresponding equation as well as putting into context with practical examples. During the tree measurements in the representative plot, measurement techniques described in the SOP (which are based on best practice), were evaluated and the PP showed a high proficiency in handling the monitoring equipment as well as sampling techniques. Further, the PP showed correctly how the example data of the representative plot (i.e., DBH) was converted to biomass (see

spreadsheet SMPP-Audit-Technical Questions-Field Visit_20220913). AENOR has reviewed the carbon calculation spreadsheets /36-47/ as well as the methodology and compared with spreadsheets from previous years to better understand the change and the update.

On the other hand, there is another methodology deviation from previous verifications and validated PD:

Module BL-ARR of VM0007 and Methodology extension AR-ACM0003 do not include specific requirements for the justification of the plantation establishment rate. To justify the conservative baseline conversion/plantation establishment rate (which is not deforestation) and to calculate leakage with a baseline deforestation rate, the PP followed Module BL-PL. The conservative approaches of BL-PL provided an analytical basis for the plantation establishment rate.

AENOR deems that the deviation meets with the criteria and specifications for permitted methodology deviations and does not negatively impact the conservativeness of the quantification of GHG emission reductions or removals.

3.3 Project Description Deviations (Rules 3.5.7 – 3.5.10)

In the current version of the Monitoring Report, there are not project description deviations. AENOR has followed the methodology, the information on the MR and the differences until getting the final version of the Monitoring Report (dated on January 2024) and confirms that the project is not looking for any project description deviation.

3.4 Minor Changes to Project Description (Rules 3.5.6)

There have been minor changes to PD which apply to the current monitoring period. The biodiversity monitoring plan has been expanded with the addition of SOPs on data collection, camera trap surveying and the use of SMART software. These changes have no impact on carbon quantification since it relates to biodiversity monitoring. None of the changes made had any effect on the project's design or compliance with CCB requirements. The minor changes to project description do not constitute significant changes in the project activities or substantial changes in the impacts of the project that are not described in validated project description, because the changes only update the biodiversity monitoring plan with more details, exceeding the minimum CCB Standards for monitoring biodiversity. Also, with the modification of the plan, the risks are decreased because of the avoiding of certain zones, which are favorable for tiger habitat.

Thus, AENOR deems that after all changes the project remains in conformance with the CCB Standards criteria and indicators and the project's validated design. Therefore, the minor changes are valid and, in any case, positive for the development of the project.

3.5 Grouped Project (G1.13 – G1.15, G4.1)

This is not a grouped project.

4 VERIFICATION FINDINGS

4.1 Public Comments (Rules 4.6)

The MR and its summary were subjected to a 30-day public comment period from 22/07/2022 to 21/08/2022. No comments were received during the public comment period. The audit team confirmed this issue against public information in VERRA database platform.

4.2 Summary of Project Benefits

Section 1 of the MR provides information about the project benefits. Achievements for the current monitoring period and for the project lifetime are detailed with specific data per categories.

Data are supported with evidence and records checked through interviews to relevant stakeholders and desk review. The section has been completed appropriately with data from the sources provided such as GIS data, records of trainings activities, stakeholders' engagement reports, employees etc.

As specific and remarkable achievements for the current monitoring period the MR in its section 1.1 states i) the hydrological function restoration by constructing 15 dams bringing the total to 203 peat compaction dams across the concession; ii) the fire prevention and fire threat reduction through held a four-day firefighting training with Manggala Agni —fire brigade under the Indonesian Ministry of Environment and Forestry, the project also supported the Village Fire Prevention Community (KMPA) from Muara Merang & Kepayang villages to attend the training and supported fire safety training for each SMPP field staff member upon their induction into the project.

In addition, the project reports in section 1.2 of MR standardized benefits related to i) GHG emission reductions, ii) forest cover, iii) training of community members, iii) employment creation, iv) improvement of livelihoods, v) health services improvement, vi) improvement of access and quality of education, vii) clean water access improvement, viii) well-being improvement, and ix) biodiversity conservation.

The audit team reviewed information reported in this section against supporting evidence listed in Appendix 1, and further information referenced in proper sections stated in standardized benefits; complementary, due to the onsite audit visit, the VVB was able to personally evidence and crosscheck all this information, the audit team has verified that all achievements reported are substantiated with information provided in the body of the CCB-VCS-PD.

In opinion of AENOR, the project benefits are credible based on the supporting documents provided by PP and evidence received during the AENOR interviews to stakeholders, records checked and field records.

4.3 General

4.3.1 Implementation Status (G1.9)

Section 2.1.1 of the MR provides the objective to be achieved by the project activities and the main tasks carried out for the monitoring period. The information is supported with additional documents, such as community impacts reports /48-171/ and biodiversity monitoring records /253-267/, that give a complete information about the achievements.

In this monitoring period, the water table level was monitored in dip wells along 22 transects up to 1 km outside of the project boundary. Data was collected at the main canal outflows and was included in hydrological modeling. The project also installed 30 camera traps (compared to 18 camera traps installed in 2020), which has increased the scope for monitoring positive biodiversity impacts.

Project technical staff obtained and analyzed satellite imagery to detect land cover change over time, such as natural regeneration of vegetation. The project land cover map is updated at each monitoring period as needed. Results of the monitoring were provided in the GIS package where the changes occurred during the monitoring period can be found. Project staff will continue to monitor the effectiveness of rewetting activities to confirm that no ecological leakage is occurring at each future monitoring event, by installing monitoring dip wells within the vicinity of canal block positions inside and bordering the project area. The project has successfully installed two SmokeD smoke detection cameras at the main camp, which are working very well. This strengthens the fire monitoring system of the SMPP project tremendously and helps to detect fires within minutes day and night. The project completed a trial for gibbon survey using AudioMoth sensors installed in five locations in the dense forest.

The PP monitors and catalog species within the project zone. Section 5 of the MR shows the biodiversity monitoring results and an assessment of net positive biodiversity impacts for this monitoring period.

The project has been registered in Indonesia's National Registration System on Climate Change (SRN PPI), the SMPP has now received a project identification number under the SRN (005-I-2020-1686) and is fully compliant with the requirements of the registration process. This register does not issue any form of environmental credit. The project is no part of an emissions trading program or any other mechanism that includes GHG allowance trading. AENOR has reviewed the information provided by the PP about the registration in the National System and confirm that the information provided is deemed correct.

Section 2.1.10 of the MR describes the contribution of the project to sustainable development goals of the Republic of Indonesia. The project activities implemented during the monitoring period have a direct impact on economic growth and development, poverty and inequality reduction, education access improvement, sustainable forest and land use and climate change mitigation and adaptation.

The implementation plan for the phased project activities has been also provided to the AENOR team along with the budget and implementation schedule. The project has achieved its objectives in Climate, Community and Biodiversity by implementing project activities in every program area as results confirm. The SMPP contributes to economic growth by providing project stakeholders with employment opportunities and a variety of livelihood enhancement activities. The project actively employs local stakeholders for forest protection and facility management and provides access to education for all project staff in addition to surrounding community members.

Section 3 of this verification report contains an exhaustive list of all deviations or changes applied to the project, including methodology deviations, project description deviations, and minor changes to the project description, validated in previous monitoring periods. AENOR deems that all deviations and changes are appropriately described, justified and supported documentation and that the project remains in compliance with the VCS and CCB rules.

In conclusion, during this verification process, AENOR has not detected project changes in regards of the project title, its purposes, and objectives. As such, the project activity accurately reflects the proposed project which is mainly focused in the following program areas: forest protection and fire prevention,

community development and peat forest ecosystem restoration. Through interviews with key staff, the auditor team confirms the main objectives of the project activity.

AENOR checked the monitoring plan contained in the validated PD and compared it with the monitoring report to verify whether there was any difference that would cause an overestimation of the GHG emission reductions in the current monitoring period. AENOR has confirmed that there are no material discrepancies between the actual monitoring system, and the monitoring plan set out in the PD and the applied methodology, except to the project deviations and changes already commented and assessed in the MR and this verification report. Also, the PP effectively monitors the required parameters to determine the project's removals by sinks and emissions by sources as required by the monitoring plan and the applicable methodology.

The parameters reported, including source, frequency and review criteria as indicated in the monitoring plan were verified to be correct and in line with the revised monitoring plan of the validated PD. Necessary management system procedures including responsibility and authority of monitoring activities have been verified to be consistent with the PD. Knowledge of personnel associated with the project activity was also found to be satisfactory. For this monitoring period there are not remaining issues from previous verification.

Tables in section 2.2.1 of the monitoring report provide complete information of activities carried out and impacts of these activities for the goals of the project. Project objectives and activities to reach them are analysed with their outputs and outcomes for the present monitoring period.

The most important milestones are described in the in section 2.2.1 of the monitoring report. The following table summarizes the assessment carried out by the audit team.

Date	Milestone(s) in the project's development and implementation	VVB assessment
January 1, 2016	Project Start Date: January 1 st , 2016.	Assessed on the validated PDD
2016	The project constructed with a permanent field camp in the project area which is now the main camp of the project. The project also constructed three protection posts along the perimeter of the project boundary.	Assessed on the validated PDD
2016 - Present	Forest Protection: The project hired full time staff that reside in project facilities within the project boundary. This marked the beginning of routine patrolling of the project area that continued through the entire monitoring period.	Assessed on the validated PDD and through review of the staff CV and direct interviews
2016 - Present	Community Engagement: The project-initiated community engagement meetings and activities in nearby villages in 2016.	Assessed on the validated PDD and through direct interviews and review of the minutes of the meetings
2016	The project began its first phase of canal blocking construction and completed five temporary box dams in the eastern part of the project area.	Assessed on the validated PDD
2017	In 2017, the project began a more extensive canal blocking program and built an	Assessed on the validated PDD

Date	Milestone(s) in the project's development and implementation	VVB assessment
	additional 75 temporary box dams throughout the project area. The project also greatly expanded its facilities by increasing the size of its office and built new living quarters, which included renovating and increasing the size of the protection posts. The project hired a full-time community development manager and additional technical staff to focus on biodiversity and forest restoration. The project also secured a long-term investment to scale up project activities in October of 2017. The project also began supporting educational programs in Muara Merang and Kepayang villages.	
July 2018	The project completed its formal boundary demarcation (<i>tata batas</i>) with the government and received approval from the government.	Assessed on the validated PDD
September 2018	The project established a tree nursery at its main camp. It also increased the size of its protection posts and installed solar panels on all the protection posts.	Assessed on the validated PDD
December 2018	The project completed a peat depth survey and developed a detailed hydrology restoration plan. It also finalized the development of VCS/CCB project document. The project built a large protection post in the eastern part of the project area to protect and monitor the project's remaining secondary forest. The project continued to expand its support of education programs in the nearby villages.	Assessed on the validated PDD
May 2019	PD (Project Documentation) completed for VCS and CCB.	Assessed on the validated PDD
July 2019	Aster Global began field visit audit for joint Validation and Verification.	Assessed on the validated PDD
November 2019	Audit completed.	Assessed on the validated PDD
November 2019	Water table level monitoring system installed.	Assessed on the validated PDD
December 2019	97 peat compaction dams constructed.	Assessed on the validated PDD
December 2019	First VCU issuance.	Assessed on the validated PDD
January 2020	91 peat compaction dams constructed, totaling to 188 units.	Assessed on the validated PDD

Date	Milestone(s) in the project's development and implementation	VVB assessment
February 2020	252 dip wells installed and monitored along with other weather parameters through an online project dashboard.	Assessed on the validated PDD
April 2020	Amid the wide-spread COVID-19 pandemic, community development programs continue under new health and safety guidelines. In addition, the project distributed medical supplies and equipment to four public health centers around the project area, which included PPE, face masks, sanitizers, and immune-building multivitamins.	Assessed on the validated PDD
July 2020	Anti-stunting program launched in collaboration with 1000 Days Fund, an NGO with a mission to end stunting by 2030.	Assessed on the validated PDD
August 2020	A comprehensive camera trap implementation guideline was completed and installed. A total of 12 additional camera traps were installed to complement the first batch installed in 2019. Photos from the camera trap have been regularly recorded by Biodiversity staff which include sightings of a Sumatran Tiger.	Assessed on the validated PDD
October 2020	66 additional automated dip well sensors deployed utilizing IoT network technology to measure water table level in near real-time. The measurement data are easily observed through the online monitoring dashboard.	Assessed on the validated PDD
October 30, 2020	Second VCU issuance for monitoring period Jan 1 st – Dec 31 st , 2019.	Assessed on the validated PDD
November 2020	A total of 52 rainwater catchment systems successfully installed in both Muara Merang and Kepayang village. The project also continues regular monitoring and training of the anti-stunting program. A new initiative to develop programs to increase students' engagement with libraries and reading interests in general also started.	Assessed on the validated PDD
December 2020	The third verification work launched, and the project appointed Aenor Internacional, S.A.U. as the VVB.	Assessed on the validated PDD
January 2021	The project has successfully launched a new program in collaboration with <i>Taman Bacaan Pelangi</i> , a non-profit organization working to establish children's libraries in remote areas, to increase students' engagement with libraries and reading interests in general. The program comprises multiple stages of activities such as teacher training, library renovation, supplying	Verification team has reviewed the evidence provided by the PP and crosschecked with direct information during the onsite visit.

Date	Milestone(s) in the project's development and implementation	VVB assessment
	children's books, and parents engagement workshop.	
February 2021	The project completed the installation of additional 15 dams, bringing the total number of permanent canal blocking constructed since 2019 to 203 unit.	Verification team has reviewed the evidence provided by the PP and crosschecked with direct information during the onsite visit.
February 2021	The project recorded the presence of the Sunda Pangolin in SMPP, which is a critically endangered species and one of the most highly trafficked species.	Verification team has reviewed the evidence provided by the PP and crosschecked with direct information during the onsite visit.
February 2021	The project has successfully installed two SmokeD smoke detection cameras at the main camp, which are working very well. This strengthens the fire monitoring system of the SMPP project tremendously and helps to detect fires within minutes day and night. A total of almost 20 kWp photovoltaic solar panel system installation has been completed to practically avoid the use of fossil fuel for the main camp and forest protection posts electricity needs.	Verification team has reviewed the evidence provided by the PP and crosschecked with direct information during the onsite visit.
April 2021	The project held a four-day fire prevention training with Manggala Agni —fire brigade under the MoEF. The training was also attended by the Village Fire Prevention Community from both Muara Merang & Kepayang villages by the project's invitation.	Verification team has reviewed the evidence provided by the PP and crosschecked with direct information during the onsite visit.
September 2021	The project completed a trial for gibbon survey using AudioMoth sensors installed in five locations in the dense forest. The gibbon song data will then be analyzed in the accompanying software to estimate the number of groups/individuals of the Gibbon.	Verification team has reviewed the evidence provided by the PP and crosschecked with direct information during the onsite visit.
September 2021	The 2020 MR verification audit has been successfully completed and the verification report has been published and uploaded to the Verra registry.	Assessed on the Verra webpage.
December 21, 2021	Third verification issued for monitoring period Jan 1 st – Dec 31 st , 2020.	Documentation assessed with the corresponding 2021 dates.

The project has not participated nor been rejected under any other GHG programs. GHG emission reductions or removals generated by the project are not included in an emission trading program or any other mechanism that includes GHG allowance trading. The project has not received or sought any other form of environmental credit. Neither has become eligible to do so since previous verification, this is the first one.

Hence, after a complete review of the different documents provided and the interviews carried out, AENOR is able to confirm that the project implementation is in accordance with the project description contained in

the PD and the implementation status described in the MR. There are not material discrepancies between project implementation and the project description.

4.3.2 Risks to the Community and Biodiversity Benefits (G1.10)

Section 2.2.6 of the MR addresses the risks to the project benefits, which are extensively described in Annex 1 of the MR. The PP has identified natural and human-induced risks. Among the first is the geological risk of earthquakes. All staff are trained in emergency response, evacuation and first aid measures in the event of a significant geological event. The risk of naturally induced fires and other risks, such as pests, extreme weather events, all of them considered as low risks for the project. Hydrology management through monitoring water table levels using dip wells throughout the project area, and rain fall measurements through several weather stations, further reduced natural and human caused risks to the project area.

In addition, the PP has developed Non-Permanence Risk Reports to estimate the risks on Climate benefits in accordance with the VCS AFOLU Non-Permanence Risk Tool v 4.0.

As stated, the changes in the PDD are minimum, and the audit team has reviewed both, the information provided in the MR and the information provided in the validated PDD. Complementary, an exhaustive assessment of the NPRR /175/ has been carried out, analysing the complementary information, and attached evidence to the NPRR. During the visit, the audit team asked about the potential risks of the project, and the interviewees mentioned that the project staff explained them all the possible risks and there is also a health and safety protocol /234/ which they follow during field work. The audit team has also reviewed this policy, which is group of rules to follow for the proper development of project activities.

AENOR deems that the PP correctly identified the risks to the project benefits but the most important is that created, and it is implementing actions to reduce or diminish the negative impacts of these risks in the benefits on the climate, community and biodiversity.

4.3.3 Community and Biodiversity Benefit Permanence (G1.11)

Due to the nature of the project and its remoteness from population centers, the PP does not anticipate any negative socio-economic impacts nor negative environmental impacts. The project has a 25-year IUP PAN/RAP carbon storage and sequestration license that is extendable up to 50 years.

Community development activities are designed to increase family incomes by creating opportunities for education and employment. The project is also making investments into small businesses and supporting an increase in education levels among residents in nearby settlements. By investing in the human capital of the region, the PP expects to create more possibilities for formal employment with more reliable and higher wages. The project will also promote the importance of conservation and hydrology management to nearby villages as a way build local stewardship for the landscape. Youth will be targeted to promote a shift away from exploitative land-use practices to activities in the formal economy.

Regarding biodiversity benefits permanence, the project is rewetting the peatlands through the construction of permanent canal blocks to restore the hydrology of this degraded area, which will reduce the risk of fire in the landscape allowing the natural vegetation to be restored over the life of the project.

AENOR has verified those activities through the desk review and during the on-site visit. During the on-site assessment, the following people were interviewed and asked about the benefits of the project among other topics. AENOR considers the activities correct to enhance project benefits beyond the project lifetime.

4.3.4 Stakeholder Access to Information (G3.1- G3.3)

Access to project information and project documents for stakeholder engagement is in the SMPP Field Office, in which there are printed versions of the PD and MR and associated annexes in English, as well as executive summaries in English and in Bahasa Indonesia, for public viewing. The summary in Bahasa was also provided to community leaders to make available in public places in the surrounding villages. Also, additional material is distributed in form of brochures and the SMPP community development representative serves as a focal point for community inquiries and questions on the project, along with the many other staff that are actively employed in the project.

The audit team verified and confirmed through interviews to different stakeholders that they have appropriate knowledge of the project and that they have been provided with access to the project information, including MR summaries and bulletins regarding project implementation. Also, AENOR has reviewed the printed documents in the office which are available for everyone as well as the registration of the documents in the Verra webpage.

Monitoring results were discussed throughout the duration of project monitoring in 2021, and information was continually disseminated to the communities through consultation with actively involved individuals, particularly SMPP residents, fishermen, and village leaders, who were encouraged to further share project information throughout their wider community. AENOR has interviewed people from such communities to understand and verify the process of dissemination and the VVB deems that the information stated is deemed correct and matches with the information obtained from the interviews.

The project continued to conduct follow-up meetings with key stakeholders that were identified at the beginning of the project throughout 2021. The purpose of these meetings was to continue building trust and fostering relationships with project stakeholders, and to engage in further efforts towards understanding local history, livelihoods potential and any community concerns. The project publicized a variety of informational materials in both Bahasa Indonesia and English using a combination of promotional materials such as letters, signs, banners, and announcements at community meetings and workshops. AENOR has reviewed these advertisements and also, during the onsite visit, interviewed several people who confirmed the possibility to participate in such informational meetings.

According to the costs, risks and benefits, during the several meetings carried out on the stakeholder consultation meetings and discussions performed in 2021 with some communities, the PP has informed them about the potential costs, risks and benefits, as well as through these consultations, the project has been able to implement programs that are based on the needs of community members and stakeholders. AENOR by interviewing these communities' representatives gathered truly information about costs risks and benefits and deems that the information of the MR is reliable.

Finally, project staff will inform relevant stakeholders in advance about the details of the audit process and prepare local groups with ample time for a potential remote meeting or in person consultation. This will ensure, stakeholders can share their feedback with the auditor regardless of logistical or travel limitations. AENOR has also asked about the auditor visit information to the interviewees, and all of them recognized to be aware of the visit of the VCS CCB auditor as well as the verification process.

4.3.5 Stakeholder Consultation (G3.4 – G3.5)

According to the PP, the project has conducted a combination of formal stakeholder consultations and informational meetings with the communities since 2016, and the feedback and advice provided by these parties is critical to the long-term success of the SMPP, and therefore all information gathered in these meetings feeds into the project's adaptive management plan and Environmental and Social Performance framework. Additionally, this feedback is a valuable opportunity for the project to enhance and adapt all aspects of project design and implementation according to guidance from relevant identified stakeholders.

Also, project staff consulted with local communities and other stakeholders to ensure mutual understanding regarding all project activities and potential impacts. During the 2021 monitoring period the project engaged with at three community fishermen, nine farmers, 40 school groups or other education facilitators, 24 village officials, eight managers from nearby concessions, five local women's groups, and 50 village residents. Through these consultations, the project has been able to implement programs that are based on the needs of community members and stakeholders. Community health center support and renovation, installation of rainwater catchment systems and anti-stunting programs are the result of extensive consultation with the local health workers and community members. Library support and renovation is another example of a program resulting from stakeholder consultations, specifically the school and education facilitators.

AENOR deems that the stakeholder consultation practices carried out by the PP during the monitoring period ensures the participation of all community groups and other stakeholders in the design and implementation of the project, respecting their values, customs, and institution, as well as optimizing community benefits. Based on the evidence provided and the testimonies of community members and representatives directly consulted by the audit team, AENOR considers that continuous communication with stakeholders has been properly carried out throughout the monitoring period, directly with communities and other stakeholders or through their legitimate representatives, and that this communication has been effective in allowing stakeholders to influence the project implementation. The stakeholder input has been properly documented and it is appropriately reflected in the project's documents. Also, during the onsite visit, the verification team asked for the consultation process and the participation of the different stakeholders in this process. All the interviewees answered to be part of the consultation process, being part of the workshops and being considered for the taking decisions process. The project proponent, according to the interviews, accepted suggestions provided from the stakeholders and the project design was developed with the recommendations of those stakeholders.

4.3.6 Stakeholder Participation in Decision-making and Implementation (G3.6)

The stakeholder involvement in project design as well as the stakeholder communication system is described in the validated PD and the MR. The audit team was able to verify the stakeholder's involvement through the different interviews and meetings conducted and through records of different meetings and workshops, and engagements. Community members demonstrated awareness and consent of the project's activities. Consultations ensure inclusive engagement with men, women, and marginalized groups, encouraging their participation in culturally appropriate ways to ensure that the project can accommodate a wide range of perspectives. AENOR has interviewed people from these communities and participants to the meetings and considers that they were able to participate in the decision-making process of the project and the PP listened their opinion and recommendation.

The PP has put a special emphasis in communication and engagement with fishermen and farmer communities adjacent to the project area that derive their livelihoods from within the project area. Feedback provided by the stakeholders has been used to better design and implement the community development program.

4.3.7 Anti-discrimination (G3.7)

The PP, in addition to enforcing all relevant anti-discrimination Indonesian laws and regulations, has developed and is implementing an equal opportunity policy /231-238/ in order to ensure compliance with CCB Standards and to avoid discrimination or harassment based on gender, race, religion, and sexual orientation. This policy is enforced through grievance system /33-35/ in place that provides all project employees, identified community members, and other stakeholders with a quick-response system in the event that any discriminatory actions or sexual harassment does occur, as described in section 2.3.11 of the MR. During this MP there have not been reporting of any stakeholder discrimination evidence.

During the process of the verification, the audit team did not find any evidence that the project is engaging in any form of discrimination. AENOR checked and confirms that the PP has developed specific measures to prevent discrimination and to guarantee equal opportunities for community members, including women and vulnerable and/or marginalized people. Complementary, through the onsite visit carried out by the verification team, AENOR had the opportunity to ask to the stakeholders and beneficiaries if they had knowledge of any sign of discrimination. None of the interviewees declared to know proofs of discrimination. Furthermore, among the interviewees, there were women (listed in Appendix 2) who declared to be part of the project without being discriminated.

4.3.8 Stakeholder Feedback and Grievance Redress Procedure (G3.8)

The PP has established a grievance redress procedure, described in the document Sumatra Merang Peatland Project (SMPP) Grievance Redress Mechanism /33-35/. The procedure is differentiated depending on whether the grievance is raised by a community member or other stakeholder or by project staff or contractor. In both cases, a Grievance Response Team (GRT) would be assembled with the aim to solve the grievance withing two weeks from its first communication. If a grievance is not amicably resolved after this process, it would be submitted to an unbiased third party for a formal mediation and arbitration process.

AENOR held interviews with a broad range of stakeholders during the onsite visit and confirmed that the grievance redress procedure described has been implemented and that no grievances were raised during the monitoring period (also verified by AENOR through the review of the notification that no grievance were raised in 2021 /33-35/ by an independent company). Therefore, AENOR's verification team is able to confirm the grievances procedures was applied as per validated CCB-VCS-PD and in accordance with VCS and CCB (G.3.8) requirements.

4.3.9 Worker Relations (G3.9 – G3.12)

The PP provides staff with an orientation program which includes overviews of the project's SOPs and policies with specific attention to health and safety policies, as part as the on-boarding process. The employees also receive training in the use of technology such as GPS units, computers, solar energy, IoT sensors and firefighting equipment. AENOR has reviewed the summaries of the trainings provided by the PP /251-252/ /203/ as well as confirmed such evidence and attendance to the trainings directly by interviews

while the onsite audit was performed. AENOR's Verification team is able to confirm that project proponent provides orientation and training for those employed through project activities and relevant people from the communities and meet the VCS and CCB (G.3.9) requirement. Some of the trainings received by the people interviewed are firefighting, wildlife conflicts or stunting prevention among others.

The MR section 2.3.14 describes the policy for human resources /236/. The project as an equal opportunity employment policy and is committed to engage with the identified community groups and local stakeholders on employment opportunities. To achieve this, the project has a policy to hire 80% of its staff from local areas as a way to increase community and stakeholder benefits. Further, the project has a target to have 25% of its work force be women to ensure greater gender balance in fields traditionally dominated by men. During this monitoring period, the project directly employed 12 women as project staff (19%). Job opportunities are publicly advertised in community sign boards and announcements via community leaders. Meetings are convened within project stakeholder villages and settlements as part of the recruitment process in order to spread the word and attract candidates.

The rights and obligations of workers are observed and enforced in accordance with Law No. 13/2003 regarding Manpower. This is an umbrella law on labor relationship in Indonesia, which stipulates the working conditions and rights and obligations of the employees and companies. In addition, the project has its own human resource policies containing the details of employment practices that are followed. These policies are explained to all staff during their orientation program and are available for review in the project's field office and posted at all forest patrol offices. During the onsite visit, the audit team asked to the interviewees if they were aware of their rights and obligations as SMPP staff, and all of them declared to be aware of the Law.

In relation to occupational risks, project staff are exposed to several risks including strenuous activities in wet, humid, and hot conditions, forest fires, transportation accidents, interactions with illegal loggers, encounters with dangerous wildlife, and exposure to mosquito borne sicknesses. Project health and safety training addresses these risks /234/ and trains staff on the proper handling of equipment and other materials. Through this training and further explanations, the PP is informed about the risks of the project and how to minimize or avoid them. Other information provided by the PP are both, the Guidelines for Reducing Risk and Impact of Dangerous Wildlife Encounters and the SOP of Forest Patrols. In the first one, which is a document to explain the risks of wildlife, specifically with tigers, the PP provides information about the considerations of this animal, the behaviors and how to react in case of any attack. There is also information about snakes' encounters, sun bear's encounters and crocodile's encounters. Regarding each animal, the workers need to take some measures, behaviors and PPE.

The other document is the Standard Operative Procedure for the forest patrols, which sets out the operational control for the SMPP forest patrols. Among the measures, the PP explains the procedure if illegal logging or hunting activities are detected, as well as the summary of steps to follow in the case of any potential danger with wildlife, which is complemented with the previous guidelines.

AENOR has reviewed the information of the MR and crosschecked with the evidence provided, the SOP and the Guidelines, and did not detect incompliance while checking the documents provided and interviewing to the workers. Thus, the audit team deems that the project fulfills with CCB requirements related to labor relations.

4.3.10 Management Capacity (G4.2 – G4.3)

The MR shows (2.4.1 Required Technical Skills and Expertise) that the technical skills of the PP and other partner organizations were maintained and that project activities were implemented successfully. Both PPs, PT GAL and Forest Carbon, have proven experience and capacity in the development and management of AFOLU project, including experience in other VCS/CCB projects.

In addition to the technical skills provided by the PP, the project has partnered with Mirova Natural Capital (previously known as Althelia Climate Fund), which supports project investments and VCUs marketing.

CVs from key personal were provided to the audit team and reviewed by AENOR /218-229/. It is the opinion of the audit team that they have the suitable and appropriate technical and management capacity to develop the project.

The MR states that both PP, PT GAL and Forest Carbon, are financially stable companies, capable of implementing and maintaining a project of the scale of the project. Additionally, the initial project activities are fully supported by an investment from Althelia Climate Fund. The verifiers were able to confirm these statements with the updated budget and cash flow worksheet /230/ and financial statements provided by the PP.

The PP states to be committed to upholding a high level of integrity and professionalism throughout all aspects of project design and implementation and that the project is fully compliant with the US Foreign Corruption Practice Act (FCPA). No evidence of any form of corruption or illegality was found during the review of the provided evidence and the interviews by the audit team. Through the onsite visit, the verification team did not obtain any proof of corruption signs, and none of the interviewees recognized to have the knowledge of corruption signs. Complementary, AENOR has reviewed the corruption policy /235/ and deems that the information provided does match with the development of the project. Hence, AENOR confirms that the project is in accordance with the VCS and CCB requirements.

4.3.11 Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)

The following documents and information are commercially sensitive and not publicly available:

- Agreements and contracts between project proponents, other project entities, and stakeholder groups – regarding implementation of project activities.
- Financial statements

AENOR has checked the information and is able to confirm that it meets the VCS Program definition of commercially sensitive information and that it is not related to the determination of the baseline scenario, demonstration of additionality, and estimation and monitoring of GHG emission reductions and removals of the project.

4.3.12 Rights Protection and Free, Prior and Informed Consent (G5.1-G5.5)

The PP holds concession licenses for the project area. There are no permanent inhabitants of the project area. Two small farms overlap with the project license area and a small number of fishermen that fish inside or near the project area. According to the MR, the project was specifically designed, following the

identification of these customary rights holders, to ensure that no impingement upon these land rights holders or their livelihoods will occur.

Based on the information provided in the monitoring report and gathered from authorities and the PP as well as some of the interviewees during the onsite visit, AENOR can confirm that the project does not lead to the involuntary removal or relocation or property rights holders from their lands or territories and that the project protects the rights of indigenous peoples, communities and other stakeholders in accordance with the Climate, Community & Biodiversity Standards and the validated project design. AENOR has also reviewed the evidence provided to demonstrate the FPIC for farmers and fishermen and considers that the information is correct and matches with the information gathered from the onsite visit and the information of the MR as well as the CCB framework. Due to the accuracy review of Verra, one finding has been raised regarding the FPIC. It is due to the issue that some FPIC engagements were made in 2019 meanwhile the project start date was on January 1st 2016. The VVB has provided deep explanations about this topic in the PRR and the MR. Specifically, the problem lay in the irregular presence of a group of fishermen in the project area. These irregular individuals were identified and consulted once they were encountered. The PP has provided evidence of the pre-2019 meeting invitations and minutes for those meetings in local language and English /268-272/. Also, the PP provided 2017 engagement with fishermen. The evidence of engagement with potentially affected rights holders in 2019, including agreements made with these rights holders obtained prior to the construction of permanent canal blocks was provided by the PP was assessed by AENOR /279/. For the new fishermen identified in 2021, the PP has provided evidence of the meetings, and documents to demonstrate the rights of these groups within the project area /295-300/.

To have a better overlook of the total group of fishermen identified, the PP has developed a general spreadsheet which contains all fishermen engaged since the project began and the encounters with them along the life of the project. This document, together with the evidence provided clearly shows that fishermen were identified from 2016 onwards and that where relevant FPIC has been obtained before project activities that could impact rights were implemented /301/. Complementary, the PP provided a study carried out in 2018 about potential property rights holders in and around the SMPP project area. this study shows the map of possible stakeholders and the resources used for each group. The report shows that it is only fishermen's rights that have been affected by the project, and these stakeholders have been engaged in FPIC processes where relevant /302/.

AENOR has also reviewed the evidence of evaluation of project impacts from potentially affected rights holders, and evidence of the project's commitment to consultation and FPIC processes. Specifically, the minutes of a 2019 meeting with a potentially affect rights holder—a farmer named Matnur—where the farmer discusses the impacts of the project, including positive impacts from temporary canal box dams, the minutes of a 2019 meeting with a potentially affect rights holder—a fisherman named Sangkut—that discusses the impacts of the project, including some positive and some negative impacts from temporary canal box dams, and how to mitigate the negative impacts and the 2019 impact evaluation report on consultations with potentially affected rights holders. The consulted participants are several fishermen and one farmer who highlight several positive impacts of the project and a few negative impacts of the project.

In section 2.5.4 of the MR, the PP identifies the illegal activities that have historically occurred within the project area and described the actions taken to reduce them. These actions are aligned with the project activities and their implementation have been confirmed by the evidence provided by the PP and the stakeholders consulted by the audit team. The Project does not and has not benefited from any illegal activity.

4.3.13 Legal Status (G5.6)

The MR lists all the relevant national and local laws and regulations in section 2.5.6. Evidence of its fulfilment is considered complete. SMPP has been registered with the SRN since 2020, as stated in the MR where the project presents its SRN registration ID (005-I-2020-1686). AENOR did not detect during the verification process any incompliances related to laws and regulations.

4.4 Climate

4.4.1 Accuracy of GHG Emission Reduction and Removal Calculations

Procedures for quantifying the baseline emissions were conducted in accordance with the methodology VM0007 version 1.5. The verification team performed an intensive review of all input data, parameters, formulas, calculations, conversions, statistics and resulting uncertainties and output data to ensure consistency with the VCS documentation, methodology and associated tools, and the CCB-VCS-PD. Further, the verification team reproduced the calculations to ensure accuracy of the results. Inventory data, formulas, and calculations were provided by project proponents in spreadsheet format to ensure all formulas were accessible for review. The verification team recalculated subsets of the analysis to confirm correctness. If needed, project proponent also provided a step-by-step overview of select calculations to ensure the verification team understood the approach and could confirm its consistency with the methodologies and CCB-VCS-PD. Where applicable, references for analysis methods or default values were checked against relevant scientific literature for best practice.

Verification team assessed the parameters listed in section 3.1 of the final version of the monitoring report, including fixed and monitored parameters and considered that they are complete and in accordance with the applied methodology and validated CCB-VCS-PD. Verification team confirms that the emission reductions, including accuracy of spreadsheet formulae, conversions and aggregations are consistent in the using of the data and parameters. Also, the methods and formulae set out in the project description for calculating baseline emissions, project emissions and leakage have been followed.

The VVB deeply reviewed the spreadsheets calculations of the SMPP project and complementary, several meetings were performed with the project technical team to explain some clarifications about the calculations. AENOR reached a reasonable level of assurance with the direct observation of the tree sampling in the sampling plots and the subsequent data analysis and comparison. AENOR traced the calculations since the raw data (which also was compared with the values obtained in situ) until the final values of AGB or VCU (including buffer pool and NPRR).

To quantify current carbon stocks in the project area, was used the procedure defined in the methodology. Complete steps to calculate emission reduction are detailed in section 3.2 of the CCB-VCS-PD and the results derived from validate project design document are listed in section 3.2 of the CCB-VCS-MR. Verification team assessed the emission reduction calculation spreadsheet /36-47/; satellite analysis /30-31/; Project Area Map /19-23/; KML files /18/; GIS data /18-25/. Results are summarized following:

Baseline emissions

Section 3.2 of the MR and the calculation spreadsheets submitted to AENOR provide information related to the baseline emissions calculations.

AENOR has checked the calculations provided and confirms that emissions in the baseline scenario are consistent with the validated PD.

Baseline emissions, which include emissions from peat microbial decomposition, uncontrolled peat burning and water bodies in the project area under the baseline scenario, were estimated in 1,150,380 tCO₂e for the monitoring period. Baseline net emission removals under the ARR baseline scenario within the project period were estimated in 60,867 tCO₂e for the monitoring period.

The resulting net GHG emissions in the project area in the baseline scenario for the monitoring period account for a total of **1,089,512 tCO₂e**.

Calculation Project Emissions

Calculation of emissions from project activities has been determined following the monitoring plan, the methodology and the PD. The project scenario accounts the following emissions and removals:

1. Above-ground biomass (AGB);
2. Peat microbial decomposition; and
3. Water table level changes due to WRC activities.

Remote satellite data analysis shows four land cover classes in the project area: 1) 69% Open canopy tree/shrub/herbaceous mix, 2) 26% Close canopy tree/shrub/herbaceous mix, 3) 1% Secondary forest, 4) 4% open/closed canopy transition mix and 5) close to 1% the rest of classes such as water bodies, plantation forest and bare ground Remote sensing data and ground fire patrols were used for hotspot and fire occurrence screening. No fires were detected within the project area during the monitoring period.

The general equation used by the PP for the estimation of the total net GHG project emissions was:

$$GHG_{WPS-WRC} = \sum_{t=1}^{t^*} \sum_{i=1}^M (E_{peatsoil-WPS,i,t} + E_{peatburn-WPS,i,t} + E_{peatditch-WPS,i,t})$$

Where:

$GHG_{WPS-WRC}$	Net CO ₂ equivalent peat GHG emissions in the project scenario up to year t^* (t CO ₂ -e)
$E_{peatsoil-WPS,i,t}$	GHG emissions from microbial decomposition of the peat soil within the project boundary in the project scenario in stratum i in year t (t CO ₂ -e yr ⁻¹)
$E_{peatburn-WPS,i,t}$	GHG emissions due to peat burning under project scenario in stratum i in year t of each GHG (CO ₂ , CH ₄) (t CO ₂ -e yr ⁻¹)
$E_{peatditch-WPS,i,t}$	GHG emissions from water bodies within the project boundary in the project scenario in stratum i in year t (t CO ₂ ⁻¹ yr ⁻¹)
i	1, 2, 3, ... M strata
t	1, 2, 3, ... t^* time elapsed since the start of the project activity (years)

The procedures for quantifying the dynamics of carbon stock and peat losses are similar to those applied in the baseline scenario and are consistent with the estimations in the validated PD. Since not enough ARR monitoring has been carried out yet for quantification, ARR GHG removals are not considered for the current monitoring period.

The summary of the project emissions for the monitoring period are shown in the table below.

Year	CO ₂ from peat microbial decomposition (t CO ₂ -e ha ⁻¹ yr ⁻¹)	CH ₄ from peat microbial decomposition (t CO ₂ -e ha ⁻¹ yr ⁻¹)	Emissions (CO ₂ and CH ₄) from uncontrolled burning (t CO ₂ -e ha ⁻¹ yr ⁻¹)	CO ₂ -e from DOC (t CO ₂ -e y ⁻¹)	Total (t CO ₂ -e ha ⁻¹ yr ⁻¹)
2021	5,320	3,874	-	201	9,395

The total project emissions for the monitoring period in the project area were **9,395 tCO₂e**.

Calculations /36-47/ and GIS files /18-32/ were provided to AENOR. A complete description of the process, assumptions and assessments carried out by proponents is provided in the MR and complementary there is an annex to explain deeper all the information related to calculations. To support the assessment, the PP and AENOR carried out technical meetings reproducing the calculations and following the procedure with some people from different technical departments such as Carbon calculation, GIS, remote sensing or monitoring.

Calculation of Leakage

The PP uses the modules of VM0007 for leakage estimation applicable to the project. These include leakage i) from activity shifting for avoiding planned deforestation and planned degradation, ii) from displacement of pre-project agricultural activities and iii) emissions from ecological leakage (LK-ECO).

For leakage from activity shifting for avoiding planned deforestation and planned degradation, the analysis of the PP described in the MR demonstrates that there was no leakage throughout the monitoring period (since all values of $LKA_{planned,i,t}$ are negative).

The areas identified for ARR activities inside the project area did not have any pre-project agricultural activities taking place. Therefore, in accordance with Module LK-ARR and the CDM tool *Estimation of the increase in GHG emissions attributable to displacement of pre-project agricultural activities in A/R CDM project activity*, leakage from displacement of pre-project agricultural activities is considered zero.

Finally, during this monitoring period the project implemented rewetting activities. The hydrological model results of the mean annual water table levels within and outside the project area within a 1 km buffer were 8.43 cm ± 0.68 and 8.64 cm ± 0.65 (annual average ± S.E.), respectively. The water table levels inside and outside the project area were not statistically different (paired t-test: df = 364, p = 1), and showed that the WRC project activities did not influence the water table levels in the surrounding area. Therefore, the ecological leakage (LK-ECO) is zero.

Hence, the total leakage emissions for the monitoring period are 0.

Net GHG Emission Reductions and Removals

Calculation of emission reductions has been provided. Audit team has found the calculation traceable and in accordance with the applied methodology and its deviations, described in section 3.2 of this report.

The following table summarizes the estimated baseline, project and leakage and the estimated net GHG emission reductions and removals.

Year	Estimated Baseline Emissions (tCO ₂ e)	Estimated Project Emissions (tCO ₂ e)	Estimated Leakage Emissions (tCO ₂ e)	Estimated NET GHG Emission Reductions (tCO ₂ e)
2021	1,089,512	(530,198)	-	1,619,710

Therefore, the project achieved a net GHG emission reductions and removals of **1,619,710 tCO₂e** during the current monitoring period.

Finally, VCUs are calculated by removing the buffer credits. The non-permanence risk rating for this project is 14% as per calculations and demonstration of the NPRR /175/ which has also been reproduced and assessed by AENOR. Therefore, during this monitoring period (01-January-2021 to 31-December-2021), the project generated **1,392,951 VCUs for issuance and 226,759 buffer credits**.

AENOR reproduced the calculations spreadsheets /36-47/ to achieve the same results and deems they are depicted clearly and correctly in the provided sheets. The AENOR verification team was able to trace them directly from the data sources. Formulae used are in compliance with monitoring plan, PD, and methodology, as well as the default values used to determine the parameters. Thus, the net amount of VCUs to be issued is accurate and realistic. Assumptions used by PP at verification were appropriately cross-checked and assessed with requested evidence.

In order to calculate the above terms, the monitoring report details the data and parameters used during the verification process. For each of them, AENOR checked its accuracy, consistency and reliability by reproducing the spreadsheets calculations /36-47/, verifying the correctness of formulae and methods used and crosschecking the data values with sources.

AENOR carried out a deep review of the technical annex and the calculations and others provided by the PP that feed data values shown in the *MR_Tables_2021_20240119* (see Appendix 1).

AENOR verified the consistency and accuracy of each parameter detailed in sections 3.1.1 and 3.1.2 of the MR by crosschecking the information with the information in section 3.3 of the PD as well as checking values and reproducing the calculations in the spreadsheets /36-47/ and GIS package /18-32/ and did not find inconsistencies between them after the closing of CARs and CLs requested. Therefore, AENOR deems that values reported for the parameters are accurate and consistent. Information was deemed accurate and consistent taking into account sources used. Other default values used are from sources well accredited and validated at validation stage.

AENOR verified the list of parameters available at validation reported in the monitoring report and values applied (if applicable) or the references to the documents. The list is complete and in compliance with the methodology and the PD.

The data and parameters monitored and used to determine the emission reductions of the project are also detailed in section 3.1.2 of the MR. AENOR verified that list is complete and in compliance with the applicable methodology and the PD. For each parameter, the references to the tables where they are used are provided.

The parameters monitored are the following: $\Delta C_{WPS-ARR}$; ΔC_{LK-ARR} ; $GHG_{WPS-WRC}$; GHG_{LK-ECO} ; $E_{peatburn-WPS,i,t}$; $E_{proxy-CO2,i,t}$; $E_{proxy-CH4,i,t}$; A_i ; $E_{peatditch-CO2,i,t}$; $A_{ditch-WPS,i,t}$; $A_{DefPA,i,u,t}$; $C_{BSL,i}$; and $A_{RRL,forest,t}$.

In order to verify the accuracy and consistency of parameters monitored and used to calculate the avoided emissions reductions achieved for the monitoring period, the AENOR verification team reproduced table by table using the sequence established in the methodology, checking the correctness of the formulae applied and assumptions used, when applicable and that values used matched with data sources. At the same time, the verification team had to check the set of other spreadsheets /36-47/ that feed the *MR_Tables_2021_20240119* calculation spreadsheet and show data inputs for calculating the terms listed above. In addition, the whole set of spreadsheets are fed from other sources, like the GIS package and other sources/reports.

AENOR has reviewed the section 1.2 (Area of deforestation) of VMD0006, which states that the following proof must be provided to demonstrate the immediate site-specific threat of deforestation:

- Legal permissibility for deforestation: The project is located on land designated as production forest where conversion to Industrial timber plantation is legally permissible and common place. This is clearly outlined in PD Section 3.1.5.
- Suitability of project area for conversion to alternative non-forest land use: The purpose of the proxy analysis is to identify areas with the same or similar characteristics that have been subject to the same land use change identified in the baseline scenario. The baseline Identification process provides enough evidence on its own for the suitability of the project area for baseline activities. This fully legitimises the baseline and highlights the real and immediate threats to the area.
- If applicable, evidence of likely transfer of ownership to baseline agent of deforestation or class of agent of deforestation must be demonstrated by one of the following forms of evidence originating prior to the date of all evidence on pursuit of carbon finance/consideration of REDD: Not applicable.
- If government approval is required for deforestation to occur, the intention to deforest within the project area must be demonstrated by evidence: Conversion to Industrial timber plantation is an approved land use for Production Forest in Indonesia, which is the forest land use designation for SMPP. The only way this data could exist and be available to the project proponent would be if there was a specific agent of deforestation identified and if that agent was working with the project and willing to share this data. As this is not the case for SMPP these requirements are not applicable.
- Intent to deforest — intention to deforest must be demonstrated by the following form of evidence originating prior to the date of all evidence on pursuit of carbon finance/consideration of REDD:
 - Where deforestation is by an identified class of agents: A documented history (for example government data or maps) of similar planned deforestation activities by class of agents, of planned deforestation within the five years previous to without-project deforestation: As above, it has been followed the methodology to demonstrate risk of conversion from an identified class of agents. In this case the intent to deforest is demonstrated through the proxy analysis (PD section 3.2.1.2 and PD Annex 9) where the deforestation patterns caused by the class of agents in other areas with the same characteristics as SMPP to determine the deforestation rate. Additionally, a map is presented in the PD that shows the extent of plantation forest conversion in the project landscape (PD map 2), as well as the land use designation map of the MoEF land use designations showing that the project area clearly lies in the Production Forest zone (PD map 23), which further justifies

- threats to the project area. The conversion history of the pulp and paper plantations surrounding the project area can be found in the document entitled SMPP - Evidence of an imminent, planned, site-specific threat – walkthrough
- Where a specific baseline agent has been identified: Either a valid and verifiable land use management plan for deforesting the project area, or a documented history (for example government data or maps) of similar planned deforestation activities by the baseline agent of planned deforestation within the five years previous to without-project deforestation. Not applicable.

The proportion of the total parcel area planned to be deforested cannot exceed the legal mandate unless common practice in a proxy area shows that the mandates are not enforced: There are no clearly mandated annual area limits for conversion to Industrial timber plantation under Indonesian law and so the projects deforestation rate meets this requirement by default. However, in PD section 3.2.1.2.4 it is shown that the 7.85% annual deforestation rate derived from the proxy analysis is conservative and much lower than the 10% -12% targeted by plantation managers interviewed.

AENOR not only verified the spreadsheet calculation, data, and parameters but also the AENOR team could verify that the Project Proponent conducted a rigorous QC/QA procedure of its field measurements and an assessment of uncertainty. Thus, AENOR deems the Project Proponent performed good practices in this assessment and concludes that GHG removals were quantified correctly in accordance with the project description and applied methodology.

After a deep and thorough review and reproduction of calculations of tables from VM0007 as well as the mandatory modules and samples to the tracks to the other spreadsheets, AENOR deems the parameters monitored and available at validation are correct, reliable, and consistent. Information in the monitoring report is in compliance with the PD, the calculations provided and the applicable methodology. Then, the results showed in the MR are reliable, consistent, and accurate.

4.4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

The data and parameters used to determine GHG emission reductions and removals are listed in section 3.1 of the MR.

The PP has standardized data collection and processing through SOPs /203-211/ and perform multiple quality assurance and quality control (QA/QC) assessments. All data collected, recorded, stored, and reported are subject to review and approval by data analysts, data managers, the technical project lead, and the project team leader with reference to each level of data management. In order to ensure the security and traceability of data entry and QA/QC, all data in raw, intermediate, and final stages have access levels pre-defined for different types of users, as defined in the monitoring plan.

The PP uses a remote sensing for land cover changes and fire detection in the project area, in combination with onsite inspections through forest patrols. The monitoring report describes the steps followed to analyze the information. Also, AENOR has carried out technical meetings with the PP and the responsible of remote sensing to review the process of data analysis and aerial images and processing.

AENOR has verified that the monitoring plan is being implemented as the described in the project description. AENOR checked that key workers are fully involved in monitoring events (training, measuring, archiving, reporting, quality control, etc.) by interviews during the onsite visit as well as with the review of

the trainings and SOPs /203-211/. QA/QC procedures are considered strict at identifying, reviewing, and handling inconsistencies found to improve the management of the project.

Roles and responsibilities along with data management and archival system are also detailed in the monitoring report and other supported documents. The PP performs internal audits of the financial, environmental, and social elements of the project each year. On environmental and social issues, any variance from the project-specific environmental and social management plans are highlighted to management and a plan to take corrective action is developed. In the event issues arise, these are noted and documented by the project management board. The board authorizes a continuous improvement plan and progress is monitored against the plan by the project management board on all identified issues. Interviews with the project staff and inspection of data and results demonstrated that the PP possess all of the competencies required for reporting of GHG emissions reductions on accurate way.

Data presented to the audit team were clear and coherent and processing steps could be traced to the corresponding sections of the methodology and monitoring plan with transparency. The PP has provided various spreadsheet with different levels of complexity to allow the understanding of all the people who want to comprehension of the calculation.

The monitoring plan provides means for internal data review and quality control, and the data presented by the PP included the results of these internal assessments. AENOR considers that information provided is sufficiency and the quality of that information is appropriate to determine the GHG removals.

4.4.3 Non-Permanence Risk Analysis

The project utilized the non-permanence risk analysis tool, AFOLU Non-Permanence Risk Tool 4.0, to assess risk according to internal risk, external risk, natural risk, and mitigation measures for minimizing risk. The verification team reviewed the Non-Permanence Risk Report /175/ following VCS Standard v4.2 section 3.2.9 and confirmed that the project adheres to the requirements set out in the VCS AFOLU Non-Permanence Risk Tool.

At all levels, the verification team evaluated the rationale, appropriateness, and justifications of risk ratings chosen by the project proponent. Each risk factor was thoroughly assessed for conformance. A brief review of each factor is shown in the table below.

Risk factor	Risk Rating	Assessment
Internal Risks		
Project Management: It is assessed using table 1 of the VCS AFOLU Risk Tool.	-4 (total may be less than zero)	a) The project will only plant native trees that are adapted to peatland ecosystems. AENOR has reviewed the species used during the onsite visit and crosschecked with proper literature to justify that they are native. Risk rating=0 is justified. b) The project works to reduce the threat of possible encroachment by local people. The majority of the project's carbon stocks are in the peat soils. Only an industrial commercial entity would be able to achieve this. However commercial entities would have no legal right to do so and would be stopped by government authorities. Since

Risk factor	Risk Rating	Assessment
		the project has started there has been no encroachment in the project area. Risk rating=0 is justified. c) Management team includes individuals with significant experience in all skills necessary to successfully undertake project activities. AENOR has reviewed the CV /218-229/ and performed interviews with the staff to check their skills and knowledge. Risk rating=0 is justified. d) Management team maintains a presence in-country and less than one travel day from project site. The VVB has checked that the HQ of the are very close to the project area. Risk rating=0 is justified. e) The management team has extensive experience in AFOLU project design and carbon accounting under the VCS program. AENOR has reviewed the development of those participations of the PP in other projects and carbon consultancy activities. Risk rating=-2 is justified. f) The project has adopted an adaptive management plan that relies on a principle in making decisions and adjustments in response to new information and outputs of the monitoring system as the result of past management actions to improve future planning and management. AENOR by the onsite visit, reviewed the way in which the PP details the management plan. Risk rating=-2 is justified.
Financial viability: It is assessed using table 2 of the VCS AFOLU Risk Tool.	0 (total may not be less than zero)	a)-c) N/A d) The project will reach breakeven point in less than 4 years. AENOR has reviewed the cashflow on the desk review and also during the onsite visit. Risk rating=0 is justified. e)-g) N/A h) The project has secured more than 80% of funding needed to cover the total cash out before the project reaches breakeven. AENOR has reviewed the cashflow on the desk review and also during the onsite visit. Risk rating=0 is justified. i) Financial resources are available and have been secured and set aside. AENOR has reviewed the cashflow (confidential) on the desk review and also during the onsite visit. Risk rating=-2 is justified.

Risk factor	Risk Rating	Assessment
Opportunity Cost: It is assessed using table 3 of the VCS AFOLU Risk Tool.	0 (total may be less than zero)	a)-c) N/A d) The project conducted an NPV analysis for its operations covering the 47-year crediting period and compared this to an NPV analysis of the establishment of an Industrial paper pulp plantation. The project was deemed to have a higher NPV. AENOR has reviewed the NPV analysis (Confidential) Risk rating=0 is justified. e)-i) N/A
Project Longevity: It is assessed using table 4 of the VCS AFOLU Risk Tool.	15 (total may not be less than zero)	a) The project holds a concession license for the entire project area. This is valid for 25 years and is renewable. Project longevity is determined as 47 years since concession issuance was in 2013 and the project start date is 2016 with a 25 additional period granted after renewal. b)N/A Risk rating=15 is justified.
Total internal risk=11 (total may not be less than zero)		
External Risks		
Land Tenure and resources access/impact: It shall be assessed using table 6 of the Risk Tool.	2 (total may not be less than zero)	a) N/A. b) The land ownership and resource access/use rights are held by different entities as the land is owned by the government with the project having right of use. AENOR has reviewed rights during the onsite visit. Risk rating=2 is justified. c)-d) There are no disputes over land tenure or ownership of the project area in more than 5 % of the project area nor disputes over access/use rights (or overlapping rights). The VVB has performed interviews asking to the interviewees about their knowledge Risk rating=0 is justified. e) The project site is roughly 30km away from the sea and hence there are no expected impacts by the sea level. AENOR has verified this information by review of the GIS /18-32/ as well as visual inspection through the onsite visit. Risk rating=0 is justified. f)-g) N/A.
Community engagement: It shall be assessed using table 7 of the Risk Tool.	-5 (total may be less than zero)	a) N/A. b) There are 15 households who derive at least a portion of their livelihoods from the project area and live within 20km from the project boundary. More than 25% of this group have been consulted through one-on-one meetings and group

Risk factor	Risk Rating	Assessment
		discussions. AENOR has reviewed the document which evidence such consultations. Risk rating=0 is justified. c) The project generates net positive impacts on social and economic well-being of local communities is validated under the CCB Standards. AENOR considers by direct interviews that the project benefits the society and generates positive impacts on the social and economical wellbeing. Risk rating=-5 is justified.
Political Risks: It shall be assessed using table 8 of the Risk Tool.	0 (total may not be less than zero)	a-b) N/A c) Indonesia presents a score of -0.16 according to the World Bank Institute's Worldwide Governance Indicators. AENOR verified the value and reliability of source. Risk rating=2 is justified. d)-e) N/A f) The country is implementing REDD+ Readiness activities. AENOR has reviewed such activities at national level. Risk rating=-2 is justified.
Total external risks=0 (Total may not be less than zero)		
Natural risks		
Fire Risk: It shall be assessed using table 10 of the Risk Tool.	LS*M=1	The likelihood of a natural fire is less than 10 years, being insignificant to carbon stocks. This project is also implementing significant fire mitigation activities such as hydrology management, forest patrols, and early warning fire detection systems. AENOR has reviewed the evidence provided about forest fires in the country and also reviewed that the PP is implementing different trainings for all the personnel about firefighting. Thus LS= 2 is reasonable. Mitigation (M) measures: 0.5
Pest and disease outbreaks: It shall be assessed using table 10 of the Risk tool.	LS*M=0	A pest or disease outbreak would only impact above ground carbon stock and not the peat layer. Since the project's carbon stocks are primarily from the peat layer, which is the major carbon pool, this risk would not have a significant impact on the project's overall carbon stocks. The likelihood of non-permanence from pests and diseases is low, as there has never been any report of a disease or pest outbreak event in any peatland forest area in Indonesia. AENOR interviewed different people and asked about the presence of pest and diseases in the forests and non of them recognized to be aware of any type of disaster. Also, AENOR has reviewed some evidence

Risk factor	Risk Rating	Assessment
		provided by the PP to support this risk, such as the Ecological aspects of Indonesian forest plantations and the Susceptibility to Fungal Pathogens of Forests Differing in Tree Diversity Thus LS= 0 is reasonable. Mitigation (M) measures: 0.5.
Extreme weather: It shall be assessed using table 10 of the Risk tool.	LS*M=0	Droughts and floods may occur less than every ten years. Flooding is not considered a significant risk since peatland vegetation are naturally adapted to waterlogged conditions. Drop in the water table during long droughts does not lead to tree mortality but ferns and non-tree vegetation can die. However, ferns and non-tree vegetation is a very small carbon pool and is therefore not significant. AENOR has performed interviews and asked about the consequences of the extreme weather such as droughts and floods and the interviewees recognized to be a minimal risk. Thus LS= 0 is reasonable. Mitigation (M) measures: Not applicable.
Geological risks: It shall be assessed using table 10 of the Risk Tool.	LS*M=2	There isn't a significant risk from geological events on the project area's carbon stocks. Within the past 100 years 24 earthquakes with a magnitude of 4.5 or higher have been recorded within 200km of the project area. AENOR has performed interviews asking about the geological risk, and complementary, AENOR has reviewed the evidence provided, such as Probabilistic Seismic Hazard Analysis for Sumatra, the Map of Earthquakes in Indonesia 1900-2019 as well as the database of volcanic eruptions in Indonesia. Thus LS= 2 is reasonable. Mitigation (M) measures: Not applicable
Total natural risks=3		
OVERALL RISK RATING=11+0+3=14. Then an overall risk rating of 14% is considered.		

AENOR has checked that information provided in the NPRR for the monitoring period is consistent with supporting documents provided. The assumptions and justifications provided to determine the risk rating of each risk factor are elaborated and they are based on provided documents using conservative assessments. AENOR deems that information provided is consistent and appropriate from reliable sources. Thus, the overall risk rating of 14% is credible and realistic.

4.4.4 Dissemination of Monitoring Plan and Results (CL4.2)

The climate monitoring plan is available for public review at the project office. The PP provided printed summaries of the monitoring reports to project community members, other stakeholders, and beneficiaries of project activities, and provided printed copies to local government officials and posted them in community center locations and at all patrol post in the project area. The project also publicized a variety of informational materials in both Bahasa Indonesia and English through the use of a combination of promotional materials such as letters, signs, banners, and announcement at community meetings and workshops.

AENOR confirmed during the on-site visit by interviewing local stakeholders the awareness about the results of the projects, its implementation and monitoring. Through different interviews, the audit team was able to contrast and confirm the information provided in the MR according to the dissemination of the monitoring plan and results. All of the interviewees recognized to know about the process and got access to the documents and summaries of the monitoring plan. Results of the climate benefits were provided in a spreadsheet calculation. AENOR reproduced the calculation to achieve the same results, checked baseline and project emissions and leakage. Further information on the process and data checks is provided in sections above. In opinion of AENOR the monitoring plan and the results were disseminated in accordance with the validated monitoring plan.

4.4.5 Optional Gold Level: Climate Change Adaptation Measures (GL1.3)

According to the reports and studies cited in the validated PD, the geographic region of the project zone is likely to exhibit various effects of climate change with greater weather anomalies. Temperatures will increase consistently with increased duration of heat waves and dry spells. There will be a considerable shift in precipitation patterns. There will be increased frequency and intensity of heavy rainfall events.

The region is vulnerable to other weather-related disasters and climate hazards such as forest and land fires, storms, and drought that damage infrastructure and degrade ecosystems, leading to loss of life, property, ecosystem services, and livelihoods. Accordingly, longer dry season and a more intense rainy season will result in prolonged drought and more intense flooding.

Changes in land-use, temperature, and precipitation may cause widespread wildfires in the project zone. Altered biophysical features, as well as vegetation composition and structure impact biodiversity and ecosystem services functioning. Climate change would exacerbate this problem. Climate change has food security and economic growth implications. Changing precipitation patterns, evaporation, run-off, and soil moisture would all impact agricultural production levels. In addition, reduced availability of freshwater will have implications for health and economic production processes.

The PP has design and is implementing a set of project activities that will contribute directly or indirectly to climate change adaptation. Section 3.3 of the MR lists these activities, their details, the achievements during the monitoring period and the adaptation benefits.

Due to the specific nature of the activities, their direct impacts and results are obvious. Furthermore, all the stakeholders asked by the audit team during the interviews were well aware of the adverse effects of climate change and the contribution of project activities to reduce these effects. All interviews mentioned primarily the effectiveness of fire prevention and how the adverse effects of fire (destruction of the forest and harmful smoke clouds) have been minimized since the project started, reporting no fires during the monitoring period.

Based on the evidence and testimonies, it is the opinion of the audit team that the activities implemented by the project deliver the intended impacts on climate change adaptation. Hence, AENOR deems that the measures that the project proponent has implemented to mitigate or get adaptation to the climate change are reliable, effective and are consistent according to the information provided on the MR.

4.4.6 Optional Gold Level: Climate Change Adaptation Benefits (GL1.4)

The MR lists the following project activities and their expected adaptation benefits:

- **Forest patrolling:**
 - The activity reduces fire risk, which is heightened by longer dry seasons and hot windy conditions, to surrounding ecosystems and speeds fire response time to mitigate fires before they spread by having a dedicated patrol staff.
 - The activity provides mechanisms for prevention, monitoring, reporting and response to minimize risks and damage caused by climate change.
 - The risk minimization enhances resilience to potential climate change impacts.
- **Fire Prevention and Monitoring:**
 - Fire prevention is critical to maintain and improve the health of the watershed, particularly for direct community members that source fresh water for agriculture and fishing from within the SMPP. These community members would otherwise have their livelihoods adversely impacted by climate change.
 - The activity provides mechanisms for prevention, monitoring, and rapid response to minimize fire risks and damage exacerbated by climate change.
 - The risk minimization enables healthy conditions to maintain biodiversity and ecosystem functioning, hence enhances resilience to potential climate change impacts.
- **Canal Blocking and Hydrology Management:**
 - The activity provides robust, replicable mechanisms to combat degradation of water resources and elongated dry seasons by preventing water subsidence from canal drainage and improving soil and water quality, to strengthens biodiversity and ecosystem services features.
 - The biophysical improvement prepares the site facing climate change, hence generates resilience to potential impacts.
- **Reforestation and Assisted Natural Regeneration (ANR):**
 - The activity conserves genetic diversity that could be threatened by climate change, supports assisted gene flow especially for more resilient locally adapted species, and raises awareness in the communities.
 - Knowledge transfer also supports community forests.
 - The activity also promotes habitat regeneration for important species affected negatively by forest conversion, frequent wildfires and loss of habitat connectivity.
- **Community Development:**
 - The activity diversifies communities' opportunities for livelihoods and employment, and thereby generating resilience to challenges brought by climate change.
 - The improved sanitation and water usage also enhance the well-being and community health.
 - Project hydrological interventions also improve water quality and quantity for the fisherman and farmers that rely on the project area for their livelihoods, reducing the risk of watershed degradation caused by climate change.
 - Project investments in the community health centre promote greater community resilience to climate-related health crises, such as respiratory issues linked to wildfire smoke. The activity also strengthens the community response to the global pandemic which can be linked to human pressure on ecosystems and thus some of the core drivers of climate change.
- **Biodiversity Management and Monitoring:**
 - The activity establishes a data-driven, traceable system to monitor changes in biodiversity features in response to climate change.
 - The system provides basis for predicting climate change impact on biodiversity and for developing conservation planning adjusted to address the impacts.
 - The system also monitors the occurrence of key prey for the Sumatran Tiger which has been documented at the site. Hunting grounds for the Sumatran Tiger are increasingly threatened by climate-caused wildfires and the project increases the availability of food for this trigger species.

- Cross-cutting (Adaptive management, Effective administration and human resources, financial management, legal compliance):
 - The activity ensures that the decision-making process to address climate change impacts is backed by sustainable business management and operations.
 - Therefore, a mechanism to generate business resilience to climate change is in place.

The audit team reviewed the activities during the onsite visit and was able to identify positive trends in terms of adaptation to climate change as result of the implementation of project activities. Regular patrols provided direct information of onsite situation and events that may took place on the project area. As for fire events, the remote data, patrol reports and general perception of community members interviewed shown an important reduction on their frequency and intensity. As for adaptation of the neighboring communities, all members interviewed shown considerable knowledge about climate change and its adverse effects, knowledge they claimed to be acquired in sensitization sessions of the project. Regarding biodiversity, the results of the monitoring show positive trends and good knowledge about different monitoring.

Based on this information, the audit team deems that activities implemented assist communities and biodiversity to adapt to the probable impacts of climate change.

4.5 Community

4.5.1 Community Impacts (CM2.1)

Since there are no permanent communities within the project area, only two groups of users of the project area were identified in the validated PD as community groups. The MR states in section 4.1.1 the community impacts achieved by the project during the monitoring period for those two groups:

- Fishermen and their families:
 - Reduced fires in the region.
 - Protection of and access to fishing grounds; and
 - Improved water quality within the project area will benefit fish stocks long-term.
- Small-scale farmers and their families:
 - Reduced fires in the region reduces risk of farm destruction.
 - Long-term livelihood security as the SMPP and farmers have an agreement to ensure the farmers can maintain access to their existing plots of land that overlap with the project area.
 - Provision of farming equipment.

The project proponent has provided different evidence to demonstrate the community benefits mentioned above. Some of these evidence are the trainings to the community /48-50/ and the surveys /239-250/ where also interviewed were performed between the PP and the affected groups. Also, the VVB has reviewed the minutes of the meetings with different groups /239-250/ such as farmers, fishermen or village representatives. Additionally, the audit team has verified these benefits through the onsite visit, where different interviews have been carried out and other documentation such as contracts has been shown to the verification team. Also, AENOR has reviewed the community policy /233/ in which Forest Carbon (FC) is committed to mitigating negative risks and impacts in order to maximize the positive effects of FC projects in the communities where the company works. Therefore, is the opinion of AENOR, that the assessment of impacts is accurate and reflects faithfully the project benefits in communities.

4.5.2 Negative Community Impact Mitigation (CM2.2)

The PP identified the potential negative impacts listed below and took measures to mitigate these impacts so that the project has had a net positive impact on communities.

As biodiversity returns to the project area as result of project activities, it is possible that daily activities of fishermen and farmer communities may be impacted by large wildlife such as sun bears or tigers. The PP has developed a wildlife human conflict reduction strategy /251/ and will train community members on how to reduce this risk. As confirmed by community members interviewed, no encounter of risk with wildlife has taken place for the moment.

Increased water levels from canal blocking and other peat rewetting strategies could potentially impact farmer productivity, but through in-person interviews farmer community members have expressed a sentiment that the decreased risk from fire from mitigation activities by the project outweighs any inconvenience caused by higher water levels. The biggest risk to smallholder farming assets is from wildfires, which could completely wipe out their investment. This has been confirmed by interviewing community members.

Regular engagement and communication with fishermen and farmers have been confirmed by the evidence provided /239-250/ and the interviews performed during the onsite visit.

In accordance with the reported information, the project does not result in net negative impacts on the wellbeing of the community groups. Assessment by the audit team concluded that the likelihood of net negative impacts on the well-being of the community groups is adequately addressed in the monitoring report. Mitigation actions are implemented, and negative impacts are reduced to the minimum.

4.5.3 Net Positive Community Well-being (CM2.3)

The project was design to generate significant net positive impacts for the fishing and farming communities within the project area. PP respects the right of access and use of the fishing grounds and buys fish to support the fishermen' livelihoods. The project also offers employment opportunities to this community to those that are interested. Better hydrology management is expected to improve the water quality in the project area which will create more abundant and protected fishing grounds. As for farmers, the PP has guaranteed the land tenure for their existing plots within the concession license and removed them from the project area as regards of carbon accounting. The project also helps reducing fires throughout the wider project landscape, by deploying a fully equipped fire-fighting team and early warning system (EWS) for fires. These efforts also protect fishing grounds or farms from being destroyed.

Additionally, community development activities directed to communities of surrounding villages and settlements outside the project area (Other stakeholders), including education, health, and access to clean water programs, also benefit farmers and fishermen, since they are part of these communities as well.

The project has assisted communities to adapt to climate change, which can be seen through the improved hydrological and fire risk management. Rewetting of peat areas reduces heat stress on forests and maintains ecosystem function on which the community relies. It also acts to mitigate the severity of flooding events in the catchment.

The interviews with community members demonstrated that fishermen and farmer communities were receiving benefits they would not otherwise have received in the absence of the project. In opinion of AENOR, the claim of net positive stakeholder well-being impacts during the monitoring period is supported by evidence, is credible and reflects faithfully the project benefits to these communities. According to AENOR observations, the net impacts of the project activities are positive for both stakeholder groups compared with their anticipated well-being conditions under the without-project.

4.5.4 Protection of High Conservation Values (CM2.4)

Section 4.1.4 of the MR describes the measures applied to maintenance of the high conservation value attributes related with community. The identified communities that interact with the project rely heavily on the protection of the HCV 5 areas for fishing and small-scale farming. The other stakeholders in nearby villages benefit from the protection of the peat swamp forest and other areas within the project area for overall improved water quality.

The SMPP utilized its community development team on the ground to continuously monitor for any potential negative impacts on these HCVs in the project area during the monitoring period. In opinion of AENOR, none of the project activities have had, nor are likely to have, a negative impact on community-related HCVs. They are designed to either protect or enhance existing HCVs, as was verified by AENOR during the verification process.

As per confirmed during the on-site visit, project proponent is providing improvement for environmental services and the activities carried on during the MR have covered the basic needs of local people. Different employments and trainings have been gathered by the local communities. Therefore, verification team is able to confirm high conservation values have not been negatively affected by the project.

4.5.5 Other Stakeholder Impacts (CM3.2-CM3.3)

Section 4.2 of the MR gathered information about the positive and potential negative impacts in the offsite stakeholders. Since there are not permanent inhabitants of the project area, all community groups of the neighboring area are classified as *other stakeholders*.

The majority of the project's community development activities are directly focus on these other stakeholders. The majority of the livelihoods of these community members do not derive from the project area, so there is no negative impact as result of project activities on the project area. On the contrary, AENOR could verify that net positive community impacts from the project activities within the project area have also positively affected stakeholders not directly impacted by these activities, as is the case with health related to fires on the project area.

Section 4.2 of the MR lists the positive impacts of other stakeholders, which are based on education, health, livelihoods, employment, and access to clean water.

AENOR has reviewed the community surveys /239-250/, which have been crosschecked with the interviews performed during the onsite visit, and the audit team concluded that the likelihood of net negative impacts on the well-being of other stakeholder groups is adequately addressed in the monitoring report and the net impacts of project activities on the well-being are positive.

4.5.6 Community Monitoring Plan (CM4.1, CM4.2, GL2.2, GL2.3, GL2.5)

A plan for monitoring community was developed early in the project lifetime and successfully validated. Community monitoring plan, including the focal issue, key results, indicator or variable, data collection method, targeted community, monitored impacts, frequency of monitoring and results of the most recent monitoring, is included in section 4.3.1 of the MR. Through document review AENOR confirmed the monitoring plan is in place and monitoring is going on.

The monitoring work was done during the monitoring period, and the monitoring was undertaken by professionally trained teams. The PP has demonstrated that monitoring is able to identify positive and negative impacts on the more vulnerable people in the communities. Regular communication is kept between project staff and community stakeholder as to get feedback regarding their attitudes and expectations toward the project and other aspects of life in the community.

Verification team assessed the evidence provided /48-171/ which are mainly the pictures and videos that evidence the community participation. The pictures provided have been contrasted with the community survey report /61/ and asked during the visit to confirm dates, frequency and sampling methods used. Finally, these evidences are in accordance with the validated project design and with the procedures and systematics used in the verification event. Complementary, through the onsite visit, the audit team assessed through interviews the implementation of the monitoring activities with the different stakeholders and activities. AENOR confirms that community monitoring plan is implemented as per validated CCB-VCS-PD.

4.5.7 Community Monitoring Plan Dissemination (CM4.3)

The community monitoring plan is available for public review at the project office. The PP provided printed summaries of the monitoring reports to project community members, other stakeholders and beneficiaries of project activities, and provided printed copies to local government officials and posted them in community center locations and at all patrol post in the project area. The project also publicized a variety of informational materials in both Bahasa Indonesia and English through the use of a combination of promotional materials such as letters, signs, banners, and announcement at community meetings and workshops.

This was verified by the audit team during stakeholder interviews, in which interviewees confirmed that they were aware of the results of the monitoring results and that the PP shares them on a regular basis.

4.5.8 Optional Gold Level: Short-term and Long-term Community Benefits (GL2.2)

Not applicable. The project is not seeking Community Gold Level validation or verification at this time.

4.5.9 Optional Gold Level: Smallholder/community member Risks (GL2.3)

Not applicable. The project is not seeking Community Gold Level validation or verification at this time.

4.5.10 Optional Gold Level: Marginalized and/or Vulnerable Community Groups (GL2.4)

Not applicable. The project is not seeking Community Gold Level validation or verification at this time.

4.5.11 Optional Gold Level: Net Impacts on Women (GL2.5)

Not applicable. The project is not seeking Community Gold Level validation or verification at this time.

4.5.12 Optional Gold Level: Benefit Sharing Mechanisms (GL2.6)

Not applicable. The project is not seeking Community Gold Level validation or verification at this time.

4.5.13 Optional Gold Level: Governance and Implementation Structures (GL2.8)

Not applicable. The project is not seeking Community Gold Level validation or verification at this time.

4.5.14 Optional Gold Level: Smallholders/Community Members Capacity Development (GL2.9)

Not applicable. The project is not seeking Community Gold Level validation or verification at this time.

4.6 Biodiversity

4.6.1 Biodiversity Changes (B2.1)

The MR states in section 5.1.1 the biodiversity changes achieved by the project during the monitoring period, considering this changes as positive for biodiversity conservation. The reported trends are the following:

- Increased number of both floral and faunal species identified in the project area (compared to previous monitoring period):
 - 34 mammals (increase of 7 species from previous monitoring report)
 - 127 birds (increase of 25 species)
 - 23 reptiles (increase of 2 species)
 - 16 amphibians (increase of 5 species from 2020)
 - 28 fish (increase of 26 species from last monitoring report)
 - 39 arthropods (dragonflies and damselflies) (no increase; not surveyed)
 - 128 plant species (88 of which are trees) (increase of 27 species from last monitoring)
- Increased bird species diversity: the transects have shown and increase in the diversity of bird species
- Increased plant species diversity: the transects have shown and increase in the diversity of plant species
- Increased mammal diversity and abundance: the transects have shown and increase in the diversity of mammal species
- Increased apex predator populations: in 2020 (last MP) two different male Sumatran tigers surveyed with camera traps. However, several tiger spoors (footprints) were recorded just outside the project area during this monitoring period, meaning that tigers are likely still using the project area but were just not detected by camera traps in this monitoring period.
- Maintenance of secondary forest land cover, closed canopy land cover, and open canopy land cover:
 - Enhanced secondary forest cover (2016 = 260 ha; 2019 = 260 ha; 2020 = 288 ha; 2021 = 277 ha)
 - Increased closed canopy cover (2016 = 330 ha; 2019 = 330 ha; 2020 = 5,392 ha; 2021 = 5,872 ha)
 - Decreased open canopy cover from regrowth into closed canopy (2016 = 21,987 ha; 2019 = 21,987 ha; 2020 = 17,173 ha; 2021 = 15,756 ha)

- Open/Closed Canopy Transition Mix (2021 = 910 ha)
 - Increased aquatic species diversity: additional survey conducted during this monitoring period by a local student recoded 26 additional fish species, bringing the total number of fish species recorded in the project area to 28 species
 - Increased invertebrate diversity: It is assumed that decreased fire hotspots, increased protection from illegal human activities, more natural and stable water table depth levels, and maintained secondary forest cover and increased closed canopy cover will all contribute to positive trends in invertebrate species populations in the long-term.

The results of the project activities on biodiversity are positive, showing increase in the variety of identified species and the preservation of the forest cover since project start.

Verification team reviewed the annex 20 about biodiversity /13/. additionally, the desk review included the assessment of the information stated in the MR and crosschecked with the IUCN red list and further information such as the community interviews and pictures provided by the project proponent from the trap cameras /253-267/.

Finally, through the onsite visit, different interviews were performed and, in those interviews, questions about biodiversity improvement were done to the interviewees, gathering information about the increasing in biodiversity and the changes between the previous to the project start period.

The VVB during the visit had the opportunity to insight different species of animals such as: *Nisaetus cirrhatus*, *Acridothores javanicus*, *Helarctos malayanus*, *Rhyothemis obsolescens*. Also, some trees species such as *Eleiodoxa conferta*, *Artocarpus sp* or *Tetramerista glabra* were observed. The VVB also took photos of several species to demonstrate the presence of high levels of biodiversity, having the opportunity to contrast later with the PP the name and real presence of these species within the project.

In opinion of AENOR, information about benefits on biodiversity from project activities is accurate since is based on record taken from project stakeholders and PP, based on sources reliable and appropriate and the attribution of biodiversity changes to the project's activities is well justified.

4.6.2 Mitigation Actions (B2.3)

There were no negative impacts on biodiversity directly resulting from project activities recorded during this monitoring period. The only potentially negative impact on biodiversity recorded during this monitoring period was the degradation of several hectares in the west of the project area by the accidental expansion of an adjacent *Acacia* plantation into the project area. To remove this mistake, the expansion was stopped, and the area was replanted with native tree species.

The PP has provided in the section 5.1.2 a chronogram of the activities related to biodiversity and their impacts at different levels of duration. In this graphic there are possible mitigation actions for potential negative impacts.

Verification team during the onsite visit, confirmed that the information stated in section 5.1.2 and along the whole biodiversity section is right. Therefore, verification team concludes that the mitigation actions taken are appropriate and in accordance with validated project description.

4.6.3 Net Positive Biodiversity Impacts (B2.2)

The demonstration of a net-positive biodiversity impact over the project lifetime has been done by comparing the biodiversity without-project scenario, an Acacia plantation, with further degradation of the peatland, the remaining vegetation and the forest, with the project's current biodiversity conditions: further habitat degradation and homogenization, continued use of the canals for drainage which heavily impacts hydrology of the peatland, potential unchecked hunting and illegal logging practices, continued poor fire management practices and exacerbated drought and increased fire incidence in the area

The main project activities that produce biodiversity impacts are i) canal blocking, ii) reforestation and assisted natural regeneration, iii) fire management and iv) increased monitoring and enforcement. Many of these project activities that are effectively maintaining and supporting biodiversity in the project area are bringing about climate and community benefits as well.

Specifically, during the 2021 period, the PP constructed 15 canal dams, provided a consistent monitoring of fires with the total absence of fires, increased the planting with 2.950 nursery seedlings, improved the biodiversity monitoring as described in sections above and continued with daily patrols. The PP has provided further explanations of the consequences of these activities and how they mitigate the impacts that could occur in the without-project scenario.

Through direct interviews with stakeholders the audit team confirmed that the information stated on the MR is true, and the project activities have only positive incomes for biodiversity as well as for social sectors. Also, AENOR has reviewed several documentations remotely and confirms that the information provided is reliable and truly. The project has created benefits within the project zone that are unparalleled in comparison with the biodiversity baseline scenario. The benefits which exist within the project zone greatly outweigh the impacts of any potential unmitigated negative offsite action. Because of the project and its implemented project activities, the net effect of the project on biodiversity in and around the project zone is positive as it was demonstrated to AENOR.

4.6.4 High Conservation Values Protected (B2.4)

The project is dedicated to maintaining and enhancing biodiversity HCVs through numerous targeted project activities. The following biodiversity HCVs were identified in the project:

- HCV 1: Species diversity, i.e. Concentrations of biological diversity including endemic, rare, endangered, or threatened species that are significant at the global, regional and national levels.
- HCV 3: Ecosystems and habitats, i.e., Rare, Threatened, or Endangered ecosystems, habitats or refugia
- HCV 4: Ecosystem services, i.e. Basic ecosystem services in critical situations, including protection of water catchments and the control of fires.
- HCV 5: Community Needs, i.e., Sites and resources fundamental for satisfying the necessities of local communities or indigenous peoples (for livelihoods, health, nutrition, water etc.), identified through the engagement with these communities or indigenous peoples.

During the onsite visit, the VVB has visited all the HCV areas and crosschecked with the GIS information of their locations as well as the document description. AENOR deems that the HCV have been well identified and through the visit it was considered that they are not affected negatively by the project implementation.

4.6.5 Invasive Species (B2.5)

The project only uses species native of the project area. No invasive species have been introduced. AENOR has reviewed the species introduced in the year 2021 and deems that they are native species.

4.6.6 Impacts of Non-native Species (B2.6)

The project only uses species native of the project area. No non-native species have been introduced.

4.6.7 GMO Exclusion (B2.7)

The project did not introduce any GMO species to the project area during this monitoring period.

4.6.8 Inputs Justification (B2.8)

The project does not use any agricultural inputs, including fertilizers, chemical pesticides, or biological control agents.

4.6.9 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Actions (B3.2)

The PP has identified a potential negative offsite biodiversity impact during the current monitoring period. Certain populations of fish now living in man-made canals may have their lifecycles, feeding strategies, mating strategies negatively affected, and their canal habitats fragmented by canal dam construction. The return of riparian vegetation to canals should gradually improve the health of the wetland ecosystem. Furthermore, the dams are constructed to create natural spillways that allow for fish passage between dams when water levels are high.. Any potential net negative impact of the canal-living fish species is more than offset by the benefits of blocking the canals for the wider peatland ecosystem (particularly terrestrial species), meaning that project activities should result in clear net biodiversity benefits, both onsite and offsite. Verification team assessed the information provided on the MR through interviews during the onsite visit and deems that the information stated on the MR is reliable and right. Complementary, AENOR has reviewed the dams to verify the spillways mentioned above.

4.6.10 Net Offsite Biodiversity Benefits (B3.3)

The project has created benefits within the project zone that are unparalleled in comparison with the baseline scenario had the project not been present. The benefits which exist within the project zone greatly outweigh the impacts of any potential unmitigated negative offsite action. As none of the potential, albeit negligible, negative biodiversity impacts require mitigation as this stage of the project, the net effect of the project on offsite biodiversity during this monitoring period is positive and should remain positive for the remainder of the project. Because of the project and its implemented project activities, the net effect of the project on biodiversity in and around the project zone is positive as it was demonstrated to AENOR.

4.6.11 Biodiversity Monitoring Plan (B4.1, B4.2, GL3.4)

A plan for biodiversity monitoring was developed early in the project lifetime and successfully validated. During this monitoring period, the project continued to implement the biodiversity monitoring plan based on the causal model for maintained or enhanced biodiversity in the project area. Through document review AENOR confirmed the monitoring plan is in place and monitoring is going on.

Some of the monitoring activities conducted during this 2021 period were camera trap surveys, point transect surveys, vegetation plot surveys and fish surveys among others. AENOR has reviewed both onsite and remotely proof of these activities, such as the installation of camera trap /253-266/ as well as some of the results of these surveys /267/.

The PP has provided a table in the biodiversity monitoring plan section that summarizes the type of monitoring, the results, the frequency, and the methods among other variables. For example, for the camera trap surveys, the potential animals to be identified are medium to large sized mammals, because these cameras have sensors that are activated when some animals cross the sensor area, and therefore, this sensor is not activated by insects for example. There are 30 camera traps during the wet season and 29 during the dry season. Each camera has surveyed for an average of 69 or 63 days (depending if it was dry or wet season), and the analysis of the information gathered with these cameras is the richness-evenness. Relative abundance and presence of species. In other tables of the MR the results of the surveys are shown and further explanation about the different methods of surveys are also explained in 5.3 sections of the MR. All the methods used for survey both flora and fauna, which are stated on the table of the BD MP, are consequently explained deeply. Also, the mentioned table, explains what methods are desirable for each specie of trigger species. For example, for mammal trigger species, as mentioned, the camera traps are the best method, while for bird trigger species, the Point Count Transect Surveys is the method used.

For the point transect survey, the design of the transects has been done according to specific literature and it is deeply explained in section 5.3.1.1. this information was crosschecked during the onsite visit, and further explained by the personnel in charge of the measurements and surveys of biodiversity.

The PP has demonstrated that monitoring is be able to identify positive and negative impacts on the biodiversity. Surveys and inventories results were provided to verifiers.

The monitoring work was done during the period from January 1st, 2021 to December 31st, 2021. The approach used for monitoring has been explained in the monitoring report.

The audit team has reviewed the evidence provided to demonstrate the biodiversity monitoring plan, as well as the pictures or videos of trigger and relevant species obtained from the visit.

Complementary, the VVB has reviewed during the onsite visit all the BD aspects and current problems associated as well as the implementation of Biodiversity monitoring plan and conservation of trigger species. The biodiversity monitoring plan complies with the validated PD.

Therefore, the monitoring plan is in compliance with the validated CCB-VCS-PD. In opinion of AENOR the monitoring plan is effective to have a real idea of the situation. Measures scheduled and designed by the project proponent to maintain or enhance the biodiversity are correct and results confirm their effectiveness

AENOR confirms dates, frequency and sampling methods used are in accordance with the validated PD and its validated minor changes and with the procedures and systematics used in the verification event. Hence, AENOR confirms that biodiversity monitoring plan is implemented as the MR and the validated PD.

4.6.12 Biodiversity Monitoring Plan Dissemination (B4.3)

Each MR is made available online through the VCS Project Database. Both an English and Bahasa Indonesia versions of the Executive summary of the PD (which includes details of the biodiversity monitoring plan) are distributed to project community members. The Executive Summary of each MR is translated into Bahasa Indonesia and given to all relevant stakeholders (including local community leaders in the project zone, adjacent concessions, and local authorities).

This was verified by the audit team during stakeholder interviews, in which interviewees confirmed that they were aware of the results of the monitoring results and that the PP shares them on a regular basis.

4.6.13 Optional Gold Level: Trigger Species Population Trends (GL3.3)

The project area and project zone have a number of endangered and critically endangered trigger species within it that qualify this project for exceptional biodiversity benefits under the CCB Standard version 3. The project area qualifies as providing exceptional biodiversity benefits by meeting the vulnerability criteria (a), which requires the regular occurrence of at least a single individual critically endangered or endangered species. 127 trigger species have been actively surveyed and identified as being present in the project area during the current monitoring period, including mammals (21), birds (39), plants (32), reptiles (7), amphibians (5), fish (18), arthropods (5).

Without the presence of the project, it is highly likely that species diversity would be negatively affected by increasing development and land-use change within the project area. Even though currently it is unlikely that the project area holds resident individuals. Rather, it is more likely that the project area is being utilized as a corridor between other land-uses in the landscape. As project activities are developed further on, the ability for these areas to support biodiversity will be strengthened and these areas will be protected. Increased forest cover, planting of tree species that are important in the lifecycle of these species, and the rewetting of drained peat areas will likely encourage population growth.

The PP has provided the name of all the trigger species in the area as well as their population trend in different situations: the without project scenario, the baseline scenario, and the project scenario. The PP has explained the measures used to enhance the conservation of these species. AENOR has reviewed the sources such as IUCN red list, to verify that the species mentioned are really threatened. Also, the VVB during the onsite visit had the opportunity to directly sight some of threatened and trigger species such as dragonflies or birds among others. Also, the PP has shown proof of these animals' population by results of the camera trap. Also, the audit team has reviewed the papers provided by the PP where there is evidence of the presence of these species within the project area. Additionally, different interviews have been performed and information about the presence of these species and their increasing, have been obtained.

The verification team is able to confirm that the presence of these species under the project area is reliable and accurate.

4.6.14 Optional Gold Level: Effectiveness of Threat Reduction Actions (GL3.4)

There has been a considerable increase in the number of species identified since the previous monitoring period. It is the opinion of the audit team that the main threats to biodiversity are being properly tackle by project activities. Habitat loss caused both for land use change and fires has been effectively reduced, as demonstrated by the data provided by the PP. The construction of canal blocking dams will contribute to

the restoration of the original ecosystem by rewetting drained peatlands. In addition, some of the community members and project staff that regularly access the project area reported during the interviews their perception of a higher presence of wildlife in comparison with the past, including birds, snakes, monkeys and bears. Also, the comparison between this MP and the previous one, has shown an increase in the trigger populations, which means that the project is working towards conservation and the results are already measurable. By the end of this monitoring period, the project had recorded a total of 127 trigger species (an increase of 32 trigger species, including 95 faunal species, and 32 floral species), which are summarized by taxonomic class in the MR.

In opinion of AENOR, the PP is taking measures that are effective at maintaining or enhancing the population status of trigger species.

4.7 Additional Project Implementation Information

There is no more additional information.

4.8 Additional Project Impact Information

There is no more additional information.

5 VERIFICATION CONCLUSION

After review of all project information, procedures, calculations, and supporting documentation and the interview process, AENOR confirms that the monitoring conducted by the Project Proponent, along with the supporting Monitoring Report, are accurate and consistent with all aforementioned Verified Carbon Standard version 4.3 and the CCB Standards Third Edition without qualifications or limitations, the validated PD, and the selected methodology (VM0007 v1.5). AENOR confirms that the Sumatra Merang Peatland Project (SMPP), Monitoring Report (Version 1.5 dated on 22/01/2024) has been implemented in accordance with the validated PD including any validated changes as applicable.

AENOR confirms all verification activities, including objectives, scope and criteria, level of assurance, monitoring and project documentation adherence to VCS Version 4 (and all associated updates), and CCB Project Design Standards (Third Edition), as documented in this report are complete. AENOR concludes without any qualifications or limiting conditions that the Sumatra Merang Peatland Project (SMPP) meets the requirements of VCS Version 4 (and all associated updates) and CCB Standards Third Edition for the monitoring period (01-January-2021 to 31-December-2021). The verification of the ex-post emissions of the Sumatra Merang Peatland Project (SMPP) has been conducted by AENOR in accordance with ISO 14064-3:2019.

The project is achieving the climate, community, and biodiversity benefits, including Gold Level Exceptional Climate and Gold Level Exceptional Biodiversity Benefits as described in the Monitoring Report.

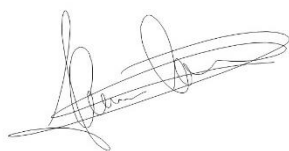
The GHG assertion provided by the project proponent and verified by AENOR has resulted in a total net GHG Emission Reductions and Removals of 1,619,710 tCO₂e by the project during the monitoring period (01-January-2021 to 31-December-2021). Considering 14% of buffer withholding based on the VCS Non-Permanence Risk Assessment Tool v4.0, which means a buffer allocation of 226,759 tCO₂e, the Verified Carbon Units (VCU) to be issued are 1,392,951 tCO₂e.

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline Emissions (t CO ₂ -e)	Project Emissions (t CO ₂ -e)	Leakage Emissions (t CO ₂ -e)	NET GHG Emission Reductions (t CO ₂ -e)
2021	1,089,215	(530,198)	-	1,619,710

Year	Net GHG emission reductions or removals(tCO ₂ e)	Non-Permanence Risk Buffer (14%)	Estimated VCUs
2021	1,619,710	226,759	1,392,951

Madrid, 22 January 2024



Javier Cócera
Lead auditor

APPENDIX 1: LIST OF EVIDENCE PROVIDED

Nº	Name of evidence
	Monitoring report
1	2021 SMPP VCS-CCB MR Master 20220610d_clean.docx
2	2021 SMPP VCS-CCB MR Master 20220610d_clean.pdf
3	2021 SMPP VCS-CCB MR Annexes Master 20220926 CLEAN.docx
4	2021 SMPP VCS-CCB MR Annexes Master 20220926 CLEAN.pdf
5	2021 SMPP VCS-CCB MR Annexes Master 20220926.docx
6	2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020 CLEAN.docx
7	2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020 CLEAN.pdf
8	2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020.docx
9	2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221108 CLEAN.docx
10	2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221108 CLEAN.pdf
11	2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221108.docx
	Monitoring report annexes
12	2021 SMPP VCS-CCB MR Annexes Master 20220610b_clean.docx
13	2021 SMPP VCS-CCB MR Annexes Master 20220610b_clean.pdf
	Monitoring report summary
14	2021 SMPP Summary Monitoring Report v1.0 English 20220610b_clean.docx
15	2021 SMPP Summary Monitoring Report v1.0 English 20220610b_clean.pdf
16	2021 SMPP Summary Monitoring Report v1.0 Indonesian 20220610a_clean.docx
17	2021 SMPP Summary Monitoring Report v1.0 Indonesian 20220610a_clean.pdf
	Supporting documents
	<u>Project location and satellite imagery</u>
18	SMPP Project Area.kml
19	Area Tanam Buring.shp
20	Area Tanam Camp.shp
21	Area Tanam Lestari.shp
22	Area Tanam Serindit.shp
23	Area Tanam Sialang.shp
24	smpp_lc_2021.shp
25	02_wtl_2021_str.shp
26	smpp_30km_buffer_hotspots_2021.shp
27	rhmc_clearing_polygons.shp
28	dipwells_20220310c_stratum.shp
29	closed_canopy_ikhb.shp
30	AGBD_2021_resample.tif
31	01_smpp_lc_2021_100m_extract_by_mask.tif
32	SMPP_wood_density_summary.xlsx

	<u>GRM</u>
33	2022 FEB-No Grievance Statement.pdf
34	Grievance Redressal Mechanism - EN.doc
35	Grievance Redressal Mechanism - ID.doc
	<u>Carbon calculation</u>
36	Emission Factors_20191101.xlsx
37	MR_Tables_2021_20220609.xlsx
38	MR_Tables_report2019_20200826.xlsx
39	Spreadsheet for ARR calculations_20190924.xlsx
40	Spreadsheet for Baseline CO2-CH4 emission_20211127.xlsx
41	Spreadsheet for MR Table 16 (burn_LUC_end_summary_area)_20220204.xlsx
42	Strata Change Summary, v9A.xlsx
43	Uncertainty_20220325.xlsx
44	R scripts
45	MR_Tables_2020_20210524.xlsx
46	MR_Tables_2021_20220923.xlsx
47	PD_Tables_20191108.xlsx
	<u>Community impacts</u>
48	2021 SMPP Anti Stunting Training.docx
49	2021 SMPP Library Training.docx
50	2021 SMPP PAH Training.docx
51	img20220330_18595242.pdf
52	amin 3.JPG
53	DSC01919.JPG
54	DSC01967.JPG
55	HapeHillmn (1).JPG
56	HapeHillmn (7).JPG
57	IMG_0137.JPG
58	IMG_20210527_155809_805.jpg
59	IMG-20210603-WA0004.jpg
60	IMG-20210603-WA0005.jpg
61	Pelatihan Metode Survey Primata_2021.pptm
62	BUKU PANDUAN MSP Ke-8 tahun 2021.pdf
63	20210123 MoM-SDN kepayang Literacy prog prep.docx
64	20210209 Library progress monitor.docx
65	20210210 Library progress monitor.docx
66	20210220 Library progress monitor.docx
67	20210222 Library Launching Merang.docx
68	20210223 Library Launching Keparang.docx

69	20210224 Parent Engagement Merang.docx
70	20210225 Parent Engagement Kepadang.docx
71	20210227 Library monitoring.docx
72	20210308 Library monitoring Kepadang.docx
73	20210310 Library Data Merang.docx
74	20210328 Program Paket coordination.docx
75	20210331 Library coordination.docx
76	20210401 Library coordination Kepadang.docx
77	20210405 Semester Exam.docx
78	20210414 MR Distribution to School.docx
79	20210423 KLHK visit to School.docx
80	20210424 Library Visitation.docx
81	20210426 Library Visitation.docx
82	20210519 MR Distribution.docx
83	20210624 TPA Activity.docx
84	20210630 Enroll new UT student.docx
85	20210702 Support distribution to TPA Merang.docx
86	20210726 Field Visit Pancuran.docx
87	20210731 Koordinasi Audit ke Stakeholder.docx
88	20210805 Audit program paket.docx
89	20210902 Update siswa paket.docx
90	20210913 SMP 2 Bayung Lencir.docx
91	20210917 SMPP Board visitation.docx
92	20210918 SMPP Board visitation Merang.docx
93	20210919 Mid Exam Monitoring and Graduation.docx
94	20210928 Library Monitoring.docx
95	20211007 Adiwiyata school.docx
96	20211116 PKBM Hidayah.docx
97	20211215 UT Students newdocx.docx
98	20210404 MoM Matnur.docx
99	20210522 MoM Matnur.docx
100	20210705 MoM Matnur.docx
101	20210730 MoM Kete dan Epa.docx
102	20210828 MoM Kete dan Epa.docx
103	20210830 MoM Kete dan Epa.docx
104	20210905 MoM Kete dan Epa.docx
105	20210302 MoM - Cikman Nelayan.docx
106	20210327 MoM - Tarmizi RT Buring.docx
107	20210812 MoM - Jon Buring.docx

108	20211109 MoM WM Coordination with RHM.docx
109	20210325 MoM - Ibu PKK Kepayang.docx
110	20210327 MoM - Tarmizi RT Buring.docx
111	20210414 MoM - Stakeholder MR Distribution.docx
112	20210428 MoM - Bansos Preparation.docx
113	20210505 MoM - Bansos Delivery.docx
114	20210519 MoM - Stakeholder MR Distribution.docx
115	20210710 Coordination Meeting with Kepayang Head of Village.docx
116	20210727 Field Visit Tebing Harapan.docx
117	20210731 MoM - Interview Audit.docx
118	20210805 MoM - Vaccination plan coordination.docx
119	20210811 MoM - Vaccination coordination.docx
120	20210929 MoM - Stunting coordination.docx
121	20210930 MoM - Stunting coordination Tebing.docx
122	20211018 MoM - Sanitation program plan.docx
123	20211107 MoM Coordination with Kades Kepayang.docx
124	20211119 MoM - Posyandu mobile plan.docx
125	20211201 MoM - Vaccination 3rd plan.docx
126	20211205 MoM - kepayang head visitation.docx
127	20211207 MoM - PAH Tagging.docx
128	20211214 MoM - PAH Material Transporting.docx
129	20211216 MoM - PAH Material Transporting monitoring.docx
130	20210117 MoM - Anti Stunting Training Prep.docx
131	20210123 MoM - TBP Library Project Prep.docx
132	20210224 MoM - Parent Engagement Merang.docx
133	20210225 Parent Engagement Kepayang.docx
134	20210228 MoM - Fishpond group Merang.docx
135	20210301 MoM - Parent engagements 2nd.docx
136	20210302 MoM - Kepayang Parent engagements 2nd.docx
137	20210303 MoM - Stakeholder interview PUSTU.docx
138	20210305 MoM - Stunting monitoring Merang.docx
139	20210306 MoM - Stunting monitoring Kepayang.docx
140	20210408 MoM - Joint training fire prevention.docx
141	20210423 MoM - Poskesdes Kepayang.docx
142	20210425 MoM - Obsevation Nuaran.docx
143	20210428 MoM - KMPA Merang Bansos Preparation.docx
144	20210430 MoM - KMPA Kepayang Bansos Preparation.docx
145	20210520 MoM - Stunting training.docx
146	20210624 MoM - Anti Stunting last monitoring.docx

147	20210624 MoM - Fishpond group monitoring.docx
148	20210624 MoM - UT Progress monitoring.docx
149	20210805 MoM - Coordination with Bidan Merang.docx
150	20210914 MoM - Fishpond group Merang.docx
151	20210917 MoM - Bidan Yessi Kepadang.docx
152	20210918 MoM - Bidan Aryati dan Erni.docx
153	20210928 MoM - Anti Stunting Prep.docx
154	20210929 MoM - Stunting monitoring Visit Merang.docx
155	20210930 MoM - Stunting monitoring Visit Kepadang.docx
156	20211001 MoM - Stunting Visit Pancuran.docx
157	20211012 MoM - Kelompok Wanita Tani Kepadang.docx
158	20211015 BTOR Handover Antigen Kit Pustu Merang.doc
159	20211016 MoM - Bidan Riana Tebing Harapan.docx
160	20211030 MoM - Farmer identification.docx
161	20211101 MoM Poskesdes Merang.docx
162	20211104 BTOR Handover Antigen Kit Kepadang.doc
163	20211115 MoM - Anti Stunting preparation.docx
164	20211116 MoM - Poskesdes Pancuran visitation.docx
165	20211117 MoM - Anti stunting workshop.docx
166	20211118 MoM - Posyandu Pancuran visitation.docx
167	20211119 MoM - Kepadang Anti stunting workshop.docx
168	20211204 MoM - Agreement with Bidan Merang.docx
169	20211208 MoM - Vaksinasi covid 19 prep.docx
170	20211217 MoM - Posyandu Mobile Tebing.docx
171	20211220 MoM - PAH Monitoring Tebing.docx
	<u>NPRR</u>
172	2021 SMPP-MR-VCS-Non-Permanence-Risk-Report-v4.0 202206010a.docx
173	WGI Indonesia 2016 - 2020.xlsx
174	2021 SMPP-MR-VCS-Non-Permanence-Risk-Report-v4.0 20221108.docx
175	2021 SMPP-MR-VCS-Non-Permanence-Risk-Report-v4.0 20221108.pdf
176	2021 SMPP-VCS-Risk-Report-Calculation-Tool-v4.0 20220401.xls
177	World Bank Governance Indicators Indonesia 2016 - 2020.xlsx
	<u>Benefits of the project</u>
178	2021 SMPP MR Data - Afterschool Teacher 20220928.xlsx
179	2021 SMPP MR Data - Fishermen 20220926.xlsx
180	2021 SMPP MR Data - Rainwater Catchment System 20220923.xlsx
181	2021 SMPP MR Data - University Scholarship 20220928.xlsx
	<u>Employment</u>
182	2021 SMPP MR Data - HR Dashboard 20220331.xlsx

	<u>Informational meetings</u>
183	002 Antigen Training Letter to Head of Muba Health Service.pdf
184	003 Antigen Training Letter to Head of Sumsel Province Health Service.pdf
185	004 A Pustu dan Poskesdes MRG Grant Letter.pdf
186	004 B Poskesdes KPY Grant Letter.pdf
187	004 Poskesdes KPY dan MRG Grant Letter.pdf
188	20210928 Letter to Merang Village Head PAH.pdf
189	DJI_0474.JPG
190	DSC_0114.JPG
191	IMG-20210301-WA0009.jpg
192	IMG-20211213-WA0004.jpg
193	IMG20210811125323.jpg
194	IMG20220815100755_01.jpg
195	IMG20220815150435.jpg
	<u>Vist of the auditor</u>
196	202209 Daftar Penerima Summary Monitoring Report SMPP.pdf
197	IMG-20220714-WA0001.jpg
198	IMG-20220714-WA0003.jpg
199	IMG-20220716-WA0003.jpg
200	IMG-20220805-WA0013.jpg
201	IMG20220820112840.jpg
202	IMG20220820123812.jpg
	<u>Risk mitigation</u>
203	2021 Fire Prevention Training - Manggala Agni.pdf
204	Fire Prevention and Monitoring 01.jpg
205	Fire Prevention and Monitoring 02.jpg
206	Fire Prevention and Monitoring 03.jpg
207	SMPP Forest Patrol 01.jpg
208	SMPP Forest Patrol 02.jpg
209	SMPP Forest Patrol 03.jpg
210	SOP Forest Patrol 20220622 EN.docx
211	SOP Forest Patrol 20220622 ID.docx
212	Anti-mosquito Lotion.jpeg
213	Anti-mosquito Net In Use.jpeg
214	Anti-mosquito Net packed.jpeg
215	Long sleeves uniform 1.jpg
216	Long sleeves uniform 2.jpg
217	Long sleeves uniform 3.jpg
	<u>CV or staff resumes</u>

218	Abdul Haris 201705.pdf
219	Achmad Susilo 2019.pdf
220	Agus Afijanto 2018.pdf
221	Angga Pratama 2022.pdf
222	Catur Prasetyo 2022.pdf
223	Jeffrey Chatellier 20190912.pdf
224	Mahardhitama 20190710.pdf
225	Mary Osmond 2022.pdf
226	Ophelia Wang 202206.pdf
227	Reagen Pangestu 2022.pdf
228	Sebastian Persch 2022.pdf
229	Zainuddin 2022.pdf
	<u>Financial health and cash flow</u>
230	Confidential
	<u>Policies and commitments</u>
231	1. Forest Carbon Statement on Policies - Updated 20210813.pdf
232	2. Environment Policy - Updated 20210813.pdf
233	3. Community Policy - Updated 20210813.pdf
234	4. Occupational Health and Safety Policy - Updated 20210813.pdf
235	5. Anti-Bribery and Anti-Corruption Policy - Updated 20210813.pdf
236	6. Human Resources and Labor Policy - Updated 20210813.pdf
237	7. Security Policy - Updated 20210813.pdf
238	8. Whistleblower Policy - Updated 20210813.pdf
	<u>Community consultations</u>
239	20210404 MoM Matnur.docx
240	20210522 MoM Matnur.docx
241	20210705 MoM Matnur.docx
242	20210730 MoM Matnur.docx
243	20210828 MoM Matnur.docx
244	20210830 MoM Matnur.docx
245	20210905 MoM Matnur.docx
246	20211009 MoM Matnur.docx
247	20211128 MoM Matnur.docx
248	20210302 MoM - Cikman Nelayan.docx
249	20210327 MoM - Tarmizi RT Buring.docx
250	20210812 MoM - Jon Buring.docx
	<u>Wildlife conflict training</u>
251	Human-Wildlife conflict mitigation training report_ENG.docx
252	Laporan pelatihan mitigasi konflik satwaliar dan manusia.pdf

	<u>Camera trap evidence</u>
253	SMPP_Helarctos malayanus_20211121.JPG
254	SMPP_Macaca fascicularis_20211025.JPG
255	SMPP_Macaca nemestrina_20211020.JPG
256	SMPP_Macaca nemestrina_20211103.JPG
257	SMPP_Macaca nemestrina_20211214.JPG
258	SMPP_Paradoxurus hermaphroditus_20211211.JPG
259	SMPP_Presbytis mitrata_20211124.JPG
260	SMPP_Prionailurus bengalensis_20211019.JPG
261	SMPP_Rusa unicolor_20211130.JPG
262	IMG_20210927_104811.jpg
263	IMG_20210929_111751.jpg
264	IMG_20210930_113146.jpg
265	IMG_20211102_092902.jpg
266	IMG_20211122_115132.jpg
267	2021 Camera Trap Locations Coordinates.csv
	<u>New FPIC evidence PRR</u>
268	2017 Invitation for Fishermen 20 November_ENG.docx
269	2017 Invitation for Fishermen 20 November_IND.docx
270	2017 June_Documentation of SMPP Project Area Fisherman Engagement.docx
271	2017 MoM Fishermen 26 November_ENG.docx
272	2017 MoM Fishermen 26 November_IND.docx
273	1. Agreement made between Project and Farmer Matnur_20180711.jpg
274	1. Matnur Farmer Agreements Transcript Translation_IND to ENG_20180711.docx
275	2. Results of follow up meeting with Farmer Matnur_20190724.pdf
276	2. ToR COMDEV Farmer Matnur_20190725.pdf
277	3. Delivery Receipt of Goods by Farmer Matnur_20190904.pdf
278	3. MoM of meeting between Project and Farmer Matnur_20190904.docx
279	Fish Farmer Group Establishment Letter_2019.pdf
280	Statement Letter_Fisherman Tasi_IND.pdf
281	Statement Letter_Fisherman Sangkut_IND.pdf
282	Statement Letter_Fisherman Nisi_IND.pdf
283	Statement Letter_Fisherman Marzuki_IND.pdf
284	Statement Letter_Fisherman Kuncit_IND.pdf
285	Statement Letter_Fisherman Kitum_IND.pdf
286	Statement Letter_Fisherman Jaini_IND.pdf
287	Statement Letter_Fisherman Hakim_IND.pdf
288	Statement Letter_Fisherman Eka_IND.pdf
289	Statement Letter_Fisherman Cikman_IND.pdf

290	Statement Letter_Fisherman Ali Hasan_IND.pdf
291	Fishermen Statement Letter Transcript_ENG.docx
292	List of Fishermen identified by SMPP_201904.pdf
293	Fishermen Statement Letter\1. Fishermen Meeting Attendance List 20190404_IND.pdf
294	Compensation Agreements\3. Fisherman Sangkut Agreement_20200128_IND_scan.pdf
295	MoM with Fisherman Cikman_20210302_ENG.docx
296	MoM with Fisherman Cikman_20210302_IND.docx
297	MoM with Fisherman Jon Buring and local leader Tarmizi_20210327_ENG.docx
298	MoM with Fisherman Jon Buring and local leader Tarmizi_20210327_IND.docx
299	MoM with Fisherman Jon Buring_20210812_ENG.docx
300	MoM with Fisherman Jon Buring_20210812_IND.docx
301	2021 SMPP MR Data - Fishermen 20230901.xlsx
302	Ekologika Social Scoping Report ENG.pptx
303	SMPP Project Impact Evaluation 20191004.docx
	<u>New evidence Carbon Calculations</u>
304	MR_Tables_2021_20230828.xlsx
305	PDT_max_year_end.csv
306	Spreadsheet for Baseline CO2-CH4 emission_20230828.xlsx
307	YEAR0.csv
308	Year01.csv
309	YEAR02.csv
310	YEAR03.csv
311	YEAR04.csv
312	YEAR05.csv
313	YEAR06.csv
314	YEAR07.csv
315	YEAR08.csv
316	YEAR09.csv
317	YEAR10.csv
318	YEAR11.csv
319	YEAR12.csv
320	YEAR13.csv

APPENDIX 2: STAKEHOLDERS INTERVIEWED

[illegible]

AENOR
Confia

PROJECT: SMPP

LOCATION: Base Camp

MEETING TOPIC:

DATE: 01-09-22

ATTENDANCE LIST

No	NAME & SURNAME	ROLE/PROFESSION/STAKEHOLDER GROUP	COMMUNITY	SIGNATURE
1	Catur Yurno Prastho	Biodiversity	SMPP	
2	Abdul Hasy Dhuwadi	Operasional	GAL	
3	Adi Jaya Prana	Field Monitoring	SMPP	
4	Mary Osmord	Director of Conservation	SMPP	
5	A. Smilo	Community Development	SMPP	
6	Kesari Ariwaga Caw	Lin Hui	GAL	
7	Agus Afid Fauzi	Operasional	SMPP	
8	Manda Wachigeli	Kasir	GAL	
9	J. Ash Winanti	Operator Radio	GAL	
10	Indra	thp gal	camp	
11	Susanto	Staf Kamp	KAMI	
12	Zainuddin	Credit Carbon Production	SMPP	
13	Sekelwa	Climate Security and MRV lead	SMPP	
14	Kristianan	Staf Admin Union	SMPP	
	maslina	tukang masak	GAL	
	Bosner		GAL	
	Adiawati		GAL	
	NUR Hasanah		GAL	

PAGE OF

AENOR
Confia

PROJECT: LOCATION: ATTENDANCE I
MEETING TOPIC: DATE: ATTENDANCE L

LOCATION: Village Kepayang 104°13'22.6"
DATE: 03 SEPT 2022 2°9'50.75"

No.	NAME & SURNAME	ROLE/PROFESSION/STAKEHOLDER GROUP	COMMUNITY	SIGNATURE
1	Laila Suhart, S. Pd. SD	Kepala Sekolah	Kepayang	[Signature]
2	Timan Purdianto, S. Pd.	Guru	Kepayang	[Signature]
3	Sungarsih, S. Pd.	Guru	Kepayang	[Signature]
4	Sumarni S Artonang	Guru	Kepayang	[Signature]
5	Rusnawati	Guru	Kepayang	[Signature]
6	Weliyanti	Guru	Kepayang	[Signature]
7	Tita Rosita	Guru	Kepayang	[Signature]
8	Delt kurnia	Guru	Kepayang	[Signature]
9	Jedi	Guru	Kepayang	[Signature]
10	Dutri El vera	Guru	Kepayang	[Signature]
11	Natasya	Pustakawati	Kepayang	[Signature]
12	Gulmin	Kades	Kepayang	[Signature]
13	Rampeli	Ketua RT	Kepayang	[Signature]
14	Yosef Am. Kab	Bidat Das Kepayang	Kepayang	[Signature]

PAGE 1 OF 1

Individual interviews

Name	Profession	Role with the project
Manda Wahyuli	Project Employees	Cashier
Ahmad Susilo Haji		Community Development Manager
Lucia Asih		Radio operator
Robet Nainggolan		Head of Forest Protection
Catur Yuono Prasetyr		Biodiversity manager
Yessi		Kepayang village health care centre
Aryati		Muara Merang Village
Jaini	CCB stakeholders	Fisherman hamlet 1 RTI Kepayang village
Ema Octavia		Secretary of the Village, daily worker in the palm oil field, Muara Merang Village

Darmanto		Ruber farmer
Natasha		Librarian Kepayang village
Riana Amelia		Midwife Muara Merang Village
John		Fisherman buring settlement
Rampeli	Traditional community leaders	representative of the sub unit of the village RT
Suleman		leader of the villag Kepayang Villag
Romila Ali		Secretary Tebing Harapan

Some of the activities verified onsite were the following:

- Project proponent, developers/management team, local team onsite
- Project activities conducted during the monitoring period (reporting, results and concordance with plan as per PD, etc.)
- Monitoring procedures, monitoring team and equipment and biomass plots layout and measurements
- Full review of operations and measurement records. Controls established to detect and correct any errors or omissions in monitoring parameters
- Forest conservation and implementation
- Project activities implemented
- Stakeholders' rights
- Stakeholders' identification and consultation, trainings/benefits/satisfaction
- Project conflicts, barriers or difficulties and grievances/complaints/conflicts
- Stakeholders' consultations, meetings and documents sharing
- Project communication
- Community, implementation of CCB community monitoring plan and benefits to the community
- Areas of High Conservation Value (HCV)
- Biodiversity improvement, benefits and changes

APPENDIX 3: CORRECTIVE ACTIONS AND CLARIFICATIONS REQUESTS

Corrective Action Request (CARs)

CAR ID:	01	Date: 20/09/2022
Description of CAR		
Within the section 1.1 and 1.2, according to the requirements, the total achievement column should include the sum of all the monitoring periods to date. The following errors have been found within the section 1, summary of project benefits: 1. In section 1.1, there are achievements for the current MP which are not included within the total achievement column. 2. In section 1.2, the sum of net estimated emissions removals in the project area is wrong. 3. In section 1.2, under category of training, the metric four people from Muara Merang... does not match between columns of MP and project lifetime. 4. In section 1.2, within the category employment, the 151 people total have been employed since project implementation is wrong. Neither is correct the number of women employed as a result of project activities. 5. In section 1.2, under livelihoods category, the sum of 22 community fishermen does not match with the sum of the current MP. Either the ten elementary school teachers and seven afterschool teachers which do not match with the MP. Finally, the 35 women have been directly employed by SMPP is wrong according to the other verification processes. 6. In section 1.2, within the number of women with improved livelihoods or income generated as result of project activities, the achievement during the project lifetime is the same than for the previous MP although there have been new achievements in the current MP. The same issue for the six women as afterschool program teachers. 7. In section 1.2, within health category, the total achievement does not include the current MP 8. In section 1.2, within health category, for the metric number of women for whom health... the total achievement is the same than the previous MP, meanwhile new achievements for the current MP have been identified. The same issue for the 258 total students including 12 students who received scholarships and the 82 students received scholarships to attend adult continuation and the 131 total female students. 9. In section 1.2, for the water category, the 77 rainwater catchment systems does not match with the total achievement of the previous year. 10. In section 1.2, within well-being category, the total sum for the project lifetime of the total number of community members whose wellbeing was improved as a result of project activities, as well as the number of women whose wellbeing was improved, should be reviewed.		
Country participant response		Date: 23/09/2022

1. The achievements from the current MP are all now included within the total achievement column.
2. The values for the sum of net estimated emissions removals in the project area has been corrected from 1,884,598 to 1,884,599, which is the sum of the estimated NET GHG emissions reductions from ARR activities for the years 2016-2021 (see file MR_Tables_2021_20220923, tab SBM - Project Summary, Cell D10).
3. The metric four people from Muara Merang... is now matched between columns of MP and project lifetime.
4. The number 151 refers to the total number of people directly employed by the project from its inception to the end of the current MP. The number, therefore, also includes resigned employees. In 2021, the number of active employees of the project was 63. Please refer to the file 2021 SMPP MR Data - HR Dashboard 20220331.xlsx for the list of SMPP employees up to 31 December 2021. The same explanation applies to the number of women employed.
5. The total number of recorded fishermen has been updated to 30. This is the cumulative total number of fishermen identified since the project's inception (2016). The 30 recorded fishermen also include those previously identified that are no longer fishing in the area. In 2021, the project only identified 14 remaining active community fishermen. The rest are either deceased, have moved to another area, or are working on something else. Please refer to the file 2021 MR Data - Fishermen 20220926.xlsx for the full list of fishermen identified.
The ten elementary school teachers were cumulative total number of the teachers supported financially through the university scholarship since project inception (2016), in 2021 there were only seven teachers supported. Please refer to the file 2021 SMPP MR Data - University Scholarship 20220928.xlsx for the full list of university scholarship recipients.
The total number of afterschool teachers supported financially is revised to eight. Please refer to the file 2021 SMPP MR Data - Afterschool Teacher 20220928.xlsx for the full list of the afterschool teachers supported.
6. The achievements during the project lifetime were already updated with the achievements from the current MP. The addition includes the support to the two female librarians, five female teachers from the adult education program, and two female health workers.
7. The total achievement is now updated with achievements from the current MP.
8. The approximate number covered all female residents of the villages where the health clinics are supported by the project, therefore providing a positive impact through improved health services to all women in the villages.
The 258 total students were already updated from the previous MP (224 in 2020) and include both males and females, as well as those who are no longer students in the current MP. The number of women is the subset of the number of the total students (in 2021 there were 130 females out of 255 total active students; in 2020 there were 97 females out of 224 total active students).
9. The 77 rainwater catchment system is the total number of systems installed in both villages, where 25 were installed in 2021 and 52 were installed in 2020. There were errors in the 2020 MR where one installation in the Kepayang elementary school and one in Muara Merang health facility were mentioned separately from the other 50. The number in the 2021 MR now includes installation in both household and public facility categories. Please refer to the file 2021 SMPP MR Data - Rainwater Catchment System 20220923.xlsx for the full list of the rainwater catchment system recipients.
10. The section has been reviewed and revised.

Documentation provided by the Country Participant	
MR_Tables_2021_20220923.xlsx 2021 SMPP MR Data - HR Dashboard 20220331.xlsx 2021 SMPP MR Data - Fishermen 20220926.xlsx 2021 SMPP MR Data - University Scholarship 20220928.xlsx 2021 SMPP MR Data - Afterschool Teacher 20220928.xlsx 2021 SMPP MR Data - Rainwater Catchment System 20220923.xlsx	
VVB Assessment	Date: 26/10/2022
1. The section has been updated and deemed correct. 2. The section has been updated and confirmed with supporting evidence. 3. The section has been updated and deemed corrected. 4. The section has been updated and assessed as correct with supporting evidence. 5. The section has been updated and confirmed with supporting evidence. 6. The section has been updated and supporting evidence confirmed that the number of fishermen, elementary school teachers and afterschool teachers are correct. 7. The section has been updated and deemed correct. 8. According to the MR template, the total achievement column should be the sum of all the monitoring periods. Please, update or provide further explanations. 9. The section has been updated and confirmed with supporting evidence. 10. section updated and deemed correct).	
Country participant response	Date: 08/11/2022
1. N/A 2. N/A 3. N/A 4. N/A 5. N/A 6. N/A 7. N/A 8. The health service supported by SMPP is open to all female residents in Muara Merang and Kepayang villages and has been since its establishment. Because all women in both villages have access to health services supported by SMPP the total value given is a population estimate of women in the villages. Without tracking population growth and demographics closely (which would be difficult) there is no year-on-year change in the number of women with access to health services supported by SMPP. When new government census data is available the numbers will be changed to reflect increases or decreases in population. Similarly, if support services were expanded to additional villages the totals would also be changed to reflect this in future monitoring periods. In short the total achievements between monitoring periods are not additive as the number of women with access to health support is capped at the population level. 9. N/A 10. N/A	
VVB Assessment	Date: 18/11/2022

1. N/A
2. N/A
3. N/A
4. N/A
5. N/A
6. N/A
7. N/A
8. The justification is deemed correct.
9. N/A
10. N/A

Therefore, CAR 01 is closed.

CAR ID:	02	Date: 20/09/2022
Description of CAR		
According to the template the following issues have been found: 1. In section 2.1.1, <i>describe how leakage and NPRR are being monitored</i>. However, this requirement is not met. 2. In section 2.3.4, <i>demonstrate how relevant and adequate information about potential costs, risks and benefits to communities – identified using a participatory and transparent process – has been provided to communities in an understandable form and in a timely manner prior to any decision they may have been asked to make with respect to participation in the project</i>, However, this requirement is not met. 3. In section 2.3.10, <i>describe the activities and / or processes implemented to enable the effective participation, as appropriate, of all communities. Demonstrate the culture-and gender-sensitive of implementation of such actions</i>, however, these requirements are not met 4. In section 2.4.1, <i>document the maintenance to the key technical skills</i> the requirement is not met. Provide also the evidence. 5. In section 2.5.2, <i>demonstrate with documented consultations and agreements [...]</i> however, the requirement is not fulfilled. 6. In section 4.1.2, <i>explain how such actions are consistent with the precautionary principle</i>, however this requirement is not met.		
Country participant response		Date: 23/09/2022

1. Leakage monitoring is now described in Section 2.1.1 with the paragraph:
Section 3.2.3.1 describes estimation of emissions from activity shifting for avoiding planned deforestation and planned degradation (LK-ASP), by comparing estimated parameters set as the baseline in PD versus actual parameters based on annual government statistics of licensed areas, as well as comparing deforested areas estimated through an annual global tree cover loss dataset versus estimates in the PD, and leakage calculations in HTI concessions (ha). Section 3.2.3.2 describes why change in estimation of emissions from displacement of pre-project agricultural activities (LK-ARR) was zero. Section 3.2.3. describes estimation of emissions from ecological leakage (LK-ECO) by monitoring water table levels affected by rewetting activities.

Non-permanence risk monitoring is now described in Section 2.1.1 with the paragraph:
In this monitoring report the non-permanence analysis was updated using the most updated Non-Permanence Risk Tool (v4.0), as detailed in project's supporting documents. The Internal Risk has been monitored through tracking project's financial and management models, The External Risk was monitored by tracking updates in the World Bank Governance Indicators of Indonesia, policy or regulatory changes, as well as changes in community and land tenure. The Natural Risk was monitored through tracking incidents of hazard occurrences. The total Internal Risk score was 11, External Risk score was 0, and Natural Risk score was 3, totaling the overall risk rating to 14. Therefore, the combined non-permanence risk buffer for the project was determined as 14%, and was applied to both WRC and ARR net emission reductions.

2. The project now describes the following narrative in Section 2.3.4:
The project has conducted a combination of formal stakeholder consultations and informational meetings with the communities in 2016 as part of the FPIC process as mentioned in the SMPP PD (Section 2.3.7). At this time the project demonstrated how relevant and adequate information about potential costs, risks and benefits to communities was provided to communities prior to any decisioning making on their behalf as it related to project participation.
Since then, no additional stakeholders have been identified that would require FPIC to be revisited and the only impacted stakeholders remain the fishermen and the single farmer. Similarly, there has been no changes to the types and scale of project activities that would necessitate FPIC processes be repeated with the current suite of impacted stakeholders. Should any new or previously unidentified stakeholder be identified, additional participatory process will be carried out to ensure ongoing compliance with CCB standards.
3. The project now describes the following narrative in Section 2.3.10:
As per the CCB standard (G3.6 and G5.2), FPIC processes only need to be applied to project-impacted communities, in which case the project should demonstrate culture-and gender-sensitive implementation. As no newly impacted stakeholders have been identified since 2016, therefore no additional FPIC processes /activities have been required. Any future FPIC processes that may be required by CCB (if additional impacted communities are identified) will ensure gender and culture sensitive implementation of participatory processes.
Outside of the CCB requirements to demonstrate culture and gender sensitive project implementation as it relates to impacted communities, the SMPP has an established community development program in the wider project zone that targets communities and community groups defined as other stakeholders. In these endeavors the SMPP has put a strong emphasis on

inclusive, fair, and equitable engagement with programs specifically targeted at women in the form of maternal health and anti-stunting programs, and children through anti-stunting and education programs for both adults and youth (see Section 4.2). The SMPP also implements an equitable and gender sensitive hiring policy with targeted gender ratios for staff and a preference for local hires where possible (see Section 2.3.14).	
4. Section 2.4.1 is now updated to show a comparison of the project team in 2019 and in 2021. The comparison shows that key positions and personnel have been maintained since the development of the PD and moreover the SMPP has grown the project team since inception.	
5. As with points two and three above no, new impacted stakeholders have been identified and therefore no further FPIC has been required since project inception. Additional elaboration is now added to Section 2.5.2: As stated in the SMPP PD Section 2.5.1, there are no outside statutory claims in the project area as the project has officially received all statutory rights to manage the area under its PAN/RAP license as described in the SMPP PD Section 2.5.7.	
6. This has now been made explicit in the MR in Section 4.1.2 with additional narrative: All the above potential negative impacts have to date not eventuated, with no-human wildlife conflicts reported or loss of livelihoods from canal blocking documented. As of 2021 there remains a low risk that negative impacts on communities will be caused by project activities. Irrespective of the low risk to communities, ongoing monitoring and the development and dissemination of mitigation strategies in anticipation of potential future risk can be considered precautionary and is in line with the application of the precautionary principle.	
Documentation provided by the Country Participant	
2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020.docx	
VVB Assessment	Date: 26/10/2022
1. The section has been updated and deemed correct. 2. The section has been updated and deemed correct. 3. The section has been updated and deemed correct. 4. The section has been updated and deemed correct. 5. The section has been updated and deemed correct. 6. The section has been updated and deemed correct. Therefore, CAR 02 is closed.	

CAR ID:	03	Date: 20/09/2022
Description of CAR		

The following incongruences have been found within section *Project description*:

1. In section 2.1.1, there are incongruences with the statement: compared to 18 camera traps installed in 2020 and the information provided in the 2020 monitoring report.
2. In section 2.1.1, there are incongruences with the figure 4,514,231 tCO₂-e
3. In section 2.1.7, there are incongruences with the evidence provided and the following statement: The SMPP sits within the Merang-Kepayang peat dome, the largest peat swamp dome in South Sumatra, which covers more than 150,000 ha

Country participant response

Date: 23/09/2022

1. The incongruency is a result of an erroneously reported number of purchased camera traps in 2019 for the 2020 Monitoring Report. The number of camera traps should have been six, instead of five. The number of camera traps used in the project for the monitoring report of 2021 is correct.
2. The value of 4,514,231 tCO₂-e was corrected to 3,347,107 tCO₂-e now representing the actual emission reductions for this monitoring report (see file MR_Tables_2021_20220923, tab SBM - Project Summary, Cell D39).
3. The value of 150,000 ha was adjusted to 140,000 ha to reflect the value reported in Reference No. 1 of the Monitoring Report.

Documentation provided by the Country Participant

MR_Tables_2021_20220923.xlsx

VVB Assessment

Date: 26/10/2022

1. The section has been updated and deemed correct.
2. The section has been updated and confirmed with supporting evidence.
3. Reference No. 1 does not refer to the Merang-Kepayang peat dome area. However, Reference No. 2 reflects the figure 150,000 ha for the area. Please update the section with this figure.

Country participant response

Date: 08/11/2022

1. N/A
2. N/A
3. The monitoring report is now updated only citing the reference Lalan, H. P. Merang REDD Pilot Project. 2010, 1–13 (2010).

VVB Assessment

Date: 18/11/2022

1. N/A
2. N/A
3. The section has been updated and deemed correct.

Therefore CAR 03 is not closed.

CAR ID:	04	Date: 20/09/2022
Description of CAR		
The following issues have been found in section 3. Climate: 1. There is one missing parameter within section 3.1.1: Project Cover Benchmark 2. In section 3.1.2, some of the parameters did not appear in the previous MP such as: $\Delta C_{(WPS-ARR)}$, $GHG_{(WPS-WRC)}$... 3. There is an incongruence between the buffer pool stated in section 3.2.4.5 of the MR and the one calculated in the NPRR.		
Country participant response		Date: 23/09/2022
1. The parameter Project Forest Cover Benchmark is specific to the VCS module VMD0015-M-REDD, which is not used in the SMPP project. The project activities of SMPP include only ARR and WRC. 2. The parameters $\Delta C_{WPS-ARR}$ and $GHG_{WPS-WRC}$ are listed in the previous MR (page 53 and 54), however the Value monitored for $\Delta C_{WPS-ARR}$ was previously zero, because the project did not yet have data to measure the Net GHG removals from ARR. 3. The value for the NPR buffer in the MR is now changed from 468,595 to 468,594 (see the calculated value in the file MR_Tables_2021_20220923, tab Table 39, Cell D3).		
Documentation provided by the Country Participant		
MR_Tables_2021_20220923.xlsx		
VVB Assessment		Date: 26/10/2022
1. The justification is deemed correct. 2. The justification is deemed correct. 3. The value has been justified with provided evidence and has been corrected in Table 40 but remains unchanged in Table 41.		
Country participant response		Date: 08/11/2022
1. N/A 2. N/A 3. The value is now also updated in the monitoring report in table 41.		
VVB Assessment		Date: 18/11/2022
1. N/A 2. N/A 3. The table has been updated and deemed correct. Therefore CAR 04 is closed.		

CAR ID:	05	Date: 20/09/2022
Description of CAR		
The CCB and VCS Monitoring report template has a specific format for the header of the document. However, this requirement is not met in some sections where is used the PDD template.		
Country participant response		Date: 23/09/2022
The correct formatting has now been applied to the corresponding sections.		
Documentation provided by the Country Participant		
VVB Assessment		Date: 26/10/2022
Header format of the document has been corrected. Therefore, CAR 05 is closed.		

CAR ID:	06	Date: 20/09/2022
Description of CAR		
In section 5.1.1 it is stated that 128 plant species were monitored in 2021, 22 more species than the last MP. However, there is an incongruence with the data form the last Monitoring Period.		
Country participant response		Date: 23/09/2022
The number of newly monitored plant species has been updated from 22 to 27 to reflect the actual increase (MR page 127).		
Documentation provided by the Country Participant		
2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020.docx		
VVB Assessment		Date: 26/10/2022
The section has been updated and deemed correct. Therefore, CAR 06 is closed.		

CAR ID:	07	Date: 20/09/2022
Description of CAR		
There are inconsistencies between the table of land cover within the MR and the values obtained from the landcover shapefile.		

Country participant response	Date: 26/09/2022
The values for each land cover class are now updated in the MR (2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020.docx) and MR Annex (2021 SMPP VCS-CCB MR Annexes Master 20220926.docx) documents.	
Documentation provided by the Country Participant	
2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020.docx 2021 SMPP VCS-CCB MR Annexes Master 20220926.docx	
VVB Assessment	Date: 26/10/2022
Please, explain what were the modifications of the MR.	
Country participant response	Date: 08/11/2022
The following documents were modified: - 2021 SMPP VCS-CCB MR Master AUDIT EDIT 20221020 (Table 26), and - 2021 SMPP VCS-CCB MR Annexes Master 20220926 (Table 34) With the changes as follows: - Updated area calculation per land cover classes. During MR 2021 development, the team conducted visual check on Plantation Forest class using high-resolution Planet imagery dated July 2021 and observed additional plantation forest that were not identified in the land cover classification analysis. In addition, the team also observed land clearing conducted by neighboring concession and was later described in more detail in Annex 17. The team delineated these additional patches and then later updated the area calculation for all classes.	
VVB Assessment	Date: 18/11/2022
The evidence has been assessed and deemed correct. Therefore CAR 07 is closed.	

Clarifications (CLs)

CL ID	01	Date: 20/09/2022
Description of CL		
Please, provide the source and the origin of the value 1,338,569 tCO2		
Country participant response	Date: 23/09/2022	
The 1,338,569 t CO ₂ -e is the annual average emission reductions calculated in the PD (see PD_Tables_20191108.xlsx, tab Table 5, Cell B51).		
Documentation provided by the Country Participant		
PD_Tables_20191108.xlsx		
VVB assessment	Date: 26/10/2022	
The evidence provided is deemed correct. Therefore, CL 01 is closed.		

CL ID	02	Date: 20/09/2022
Description of CL		
Please, provide the following explanations: 1. In section 1.2, explain why the parameter two health workers from the two villages, is categorized within livelihoods instead of health category 2. In section 1.2, explain why within the following residents from 77 households with improved water resources = 200-300 people 3. Within biodiversity conservation of section 1.2, please explain if the 22,922 ha is a new achievement or the same than the previous verification. In the case it is a new achievement, it should be summed to the total achievement column. The same for the number of globally critically species. 4. In section 2.3.3, provide further information about the dissemination of the meetings. 5. In section 2.3.5, provide further information about the information by a VVB 6. it is stated in section 2.4.5 that the project has a zero-tolerance attitude towards corruption and unethical behavior, and is not involved in, or complicit in, any form of corruption such as bribery, embezzlement, fraud, extortion, and collusion. Please explain if there is any policy about that. 7. Explain why the value E_(peatditch-CO2,i,t) was 200 in the previous MP and 201 in this MP in section 3.1.2 8. Explain the following statement: the project has developed a wildlife-human conflict reduction strategy		
Country participant response		Date: 23/09/2022
1. The two health workers are listed under the livelihoods category because they receive monthly financial compensation from the project, to support their volunteer work. The income generated improves their livelihood. 2. The project assumes each of the 77 households in Tebing Harapan hamlet consisted of 3-4 individuals, resulting in a total of around 200-300 people. 3. The Change in the number of hectares significantly better managed by the project for biodiversity conservation, measured against the without-project scenario achievement represents the total area of the project area (22,922 ha), all of which is managed for biodiversity conservation. The number of hectares did not change in the past and will not change for the lifetime of the project. The 'Number of globally Critically Endangered or Endangered species benefiting from reduced threats as a result of project activities, measured against the without-project scenario' achievement reflects the most recent survey data on species in the project area and their IUCN conservation status. The monitoring period achievement and the project lifetime achievements		

are the same because all CR, EN, and VU species recorded in the project area have benefitted from project activities since the project began.

4. The SMPP publicized various information in 2021 using promotional materials, including signs, banners, and announcements at community meetings and workshops. Please refer to folder 2.3.3 Informational Meetings for evidence of promotional materials and photos in several locations and folder Table 4 Stakeholder Consultations for the complete minutes from all meetings conducted during the current MP, as summarized in Table 4.
5. The SMPP provides the stakeholders with information regarding the verification process by an independent VVB through face-to-face announcements during the distribution of the MR summary and other visits. Please refer to folder 2.3.5 Documentation on Audit Announcement for documentation regarding this.
6. The project proponent does have policies regarding corruption and unethical behaviors. Please see the folder 2.4.5 Policies and Commitments for the latest policies, including 5. Anti-Bribery and Anti-Corruption Policy - Updated 20210813.pdf. These policies are part of the Environmental and Social Management System (ESMS), which ensures that the policies are operationalized.
7. The values for $E_{peatditch-CO2,i,t}$ vary between the 2020 and 2021 monitoring reports because the 2021 land cover analysis showed 0.43 ha more canals than reported in the previous MR, resulting from updated land cover classification analysis. (As compared between files MR_Tables_2020_20210524.xlsx, tab 27, cell J3 and file MR_Tables_2021_20220923.xlsx, tab 31, cell J3).
8. The wildlife-human conflict reduction strategy of the project consists of regular training on Human-Wildlife Conflict Mitigation. The last training was conducted in 2020 (see file Human-Wildlife conflict mitigation training report_ENG.docx) and a refresher training is planned for 2022.

Documentation provided by the Country Participant

Folder	2.3.3	Informational	Meetings
Folder	Table	Stakeholder	Consultations
Folder	4	on	Announcements
Folder	2.3.5	Audit	Commitments
Folder	2.4.5	and	20210813.pdf
5.	Anti-Bribery and Anti-Corruption	Policy - Updated	
MR_Tables_2020_20210524.xlsx			
MR_Tables_2021_20220923.xlsx			
Human-Wildlife conflict mitigation training report_ENG.docx			

VVB assessment

Date: 26/10/2022

1. The justification is deemed correct.
2. The justification is deemed correct.
3. The justification is deemed correct.
4. Provided evidence has been assessed and the justification is deemed correct.
5. The evidence has been assessed and deemed correct.
6. The evidence has been assessed and deemed correct.
7. The evidence has been assessed and deemed correct.
8. The evidence has been assessed and deemed correct.

Therefore, CL 02 is closed.

CL ID	03	Date: 20/09/2022
Description of CL		
Please, provide the following evidences: 1. In section 2.1.10.2: In 2021 the project employed 26 staff from surrounding areas out of total 43 field staffs or around 60.5%. In addition, the SMPP is working to ensure gender balance and aims to employ 25% women as project staff. During this monitoring period, the project directly employed 19% or 12 women as project staff. The project aims to reach the target in the next monitoring period 2. In table 4, provide evidence of the consultation of the 3 fishermen and the 8 managers. 3. In section 2.3.14, provide the policy mentioned in this section as well as the evidence of 12 women employed as project staff. 4. Provide the evidence of the risk mitigation action for the following risks: forest fires, interaction with illegal loggers and mosquito exposition of section 2.3.16. 5. Provide the CVs of the team of section 2.4.2 6. Provide the financial health of section 2.4.4 7. The 25 households' rainwater catchment of section 4.2 8. SMPP has been developing a wildlife human conflict reduction strategy and has trained stakeholders of the communities on how to reduce this potential negative risk. Provide such evidence 9. In table 44, provide the evidence for the camera trap survey data 10. In section 5.1.3 provide the evidence for the 2950 nurseries seedlings 11. In section 5.3.1, provide the evidence of the 30 camera trap stations		
Country participant response		Date: 28/09/2022
1. Please refer to 2021 SMPP MR Data - HR Dashboard 20220331.xlsx for the SMPP employment data and target. 2. Please refer to the Table 4 Stakeholder Consultations folder for the complete Minutes of Meetings with the fishermen and the concession managers. 3. The policy mentioned in this section refers to the target and policy stated in section 2.3.15 of the SMPP PD. Please refer to 2021 SMPP MR Data - HR Dashboard 20220331.xlsx for the SMPP employment data and target. 4. Please refer to the 2.3.16 Risk Mitigation Action folder for the complete evidence, as requested. 5. Please refer to the 2.4.2 Staff Resumes folder for the staff CVs/resumes as requested. 6. Please refer to the 2.4.4 Financial Health folder for the complete financial statements of the current MP.		

7. Please refer to the 2021 SMPP MR Data - Rainwater Catchment System 20220923.xlsx file for the complete list of the rainwater catchment system recipients. The 25 recipients in the current MP are highlighted in light yellow.
8. The wildlife-human conflict reduction strategy of the project consists of regular training on Human-Wildlife Conflict Mitigation. The last training was conducted in 2020 [see file Human-Wildlife conflict mitigation training report_ENG.docx) and a refresher training is planned for 2022.
9. In table 44, section Mammal diversity, Justification of Change, the reference to the section on the details of the camera trap data is now referenced after the first sentence.
10. A reference to Annex 21 is now included in the monitoring report.
11. The number, locations, and results of the camera trap surveys have been provided in section 5.3.1.3.

Documentation provided by the Country Participant

2021 Folder	SMPP	MR Table	Data 4	-	HR Stakeholder	Dashboard	20220331.xlsx
2021 Folder	SMPP	MR 2.3.16	Data	- Risk	HR	Dashboard Mitigation	20220331.xlsx
Folder			2.4.2		Staff		Action Resumes
Folder			2.4.4		Financial		Health
2021 Folder	SMPP	MR	Data	- Rainwater	Catchment	System	20220923.xlsx
Human-Wildlife conflict mitigation training report_ENG.docx							

VVB assessment

Date: 26/10/2022

1. The evidence has been assessed and deemed correct.
2. The evidence has been assessed and deemed correct.
3. The evidence has been assessed and deemed correct.
4. The evidence has been assessed and deemed correct.
5. The evidence has been assessed and deemed correct.
6. The evidence has been assessed and deemed correct.
7. The evidence has been assessed and deemed correct.
8. The evidence has been assessed and deemed correct.
9. The section has been updated, however, provide some examples of raw images of trap cameras.
10. The section has been updated and deemed correct.
11. the section has been updated, however, please provide the evidence of such statements about trap cameras in section 5.3.1.3

Country participant response

Date: 08/11/2022

1. N/A
2. N/A
3. N/A
4. N/A
5. N/A
6. N/A
7. N/A
8. N/A

9. Raw camera trap images have now been supplied to the auditor in a shared folder. Please refer to folder CL03 - 2021 Camera Trap Evidence.
10. N/A
11. Photographic evidence of the camera trap installation and setup has now been provided to the auditor in a shared folder. Please refer to folder CL03 - 2021 Camera Trap Evidence.

VVB Assessment

Date: 18/11/2022

1. N/A
2. N/A
3. N/A
4. N/A
5. N/A
6. N/A
7. N/A
8. N/A
9. The evidence has been assessed and deemed correct.
10. N/A
11. The evidence has been assessed and deemed correct.

Therefore CL 03 is closed.

CL ID	04	Date: 20/09/2022
Description of CL		
According to the section 2, project description, there is one less entity involved than the previous MP and this change has not been noticed within section 2.2.3 or minor changes. Please explain that.		
Country participant response		Date: 27/09/2022
The missing entity has been added to the Monitoring Report.		
Documentation provided by the Country Participant		
VVB assessment		Date: 26/10/2022
The entity has been added in section 2.1.4. Therefore, the CL 04 is closed.		

CL ID	05	Date: 20/09/2022	
Description of CL			
Please, provide the following evidence of the NPRR and reference them within the text: 1. The project's management team includes staff with significant experience in all areas necessary to manage project activities. Resumes of staff are provided to validators upon request. 2. Provide the cashflow 3. Provide the evidence of the analysis of the NPV 4. More than 20% of this group have been consulted through one-on-one meetings and group discussions. 5. The occurrence probability of the fires			
Country participant response		Date: 30/09/2022	
1. Please refer to the 2.4.2 Staff Resumes folder for the staff resumes as requested. 2. Please refer to the 1. Cash Flow and NPV [CONFIDENTIAL] folder for the cashflow analysis as requested. 3. Please refer to the 1. Cash Flow and NPV [CONFIDENTIAL] folder for the NPV analysis as requested. 4. There are currently only 14 community fishermen and one community farmer identified as being potentially impacted by the project. Three community fishermen and one community farmer (four out of 15 or about 26.7%) have been directly engaged during the 2021 MP. Please refer to the 2. Community Consultations folder for the MoM of the consultation with this group. 5. Low fire occurrence probability is justified based on the comparison of actual fire incidents and near-real-time fire hotspot data provided by NASA FIRMS during all monitoring periods as shown in the table below.			
Year	# of hotspots (high confidence level)	# of hotspots (low to nominal confidence level)	# of actual fire incidents on site
2016	0	0	0
2017	0	0	0
2018	0	0	0
2019	1	8	1
2020	0	0	0
2021	0	0	0
Total	1	8	1

Documentation provided by the Country Participant	
Folder 2.4.2 Folder 1. Cash Flow and Staff NPV Folder 2. Community Consultations	Resumes [CONFIDENTIAL]
VVB assessment	Date: 26/10/2022
The updated NPRR has not been provided.	
Country participant response	Date: 08/11/2022
Please refer to 2021 SMPP-MR-VCS-Non-Permanence-Risk-Report-v4.0 20221108.docx under the folder CL05 - Non-Permanence Risk for the updated NPRR.	
VVB Assessment	Date: 18/11/2022
The updated NPRR has been provided and deemed correct with supporting evidence. Therefore, CL 05 is closed.	